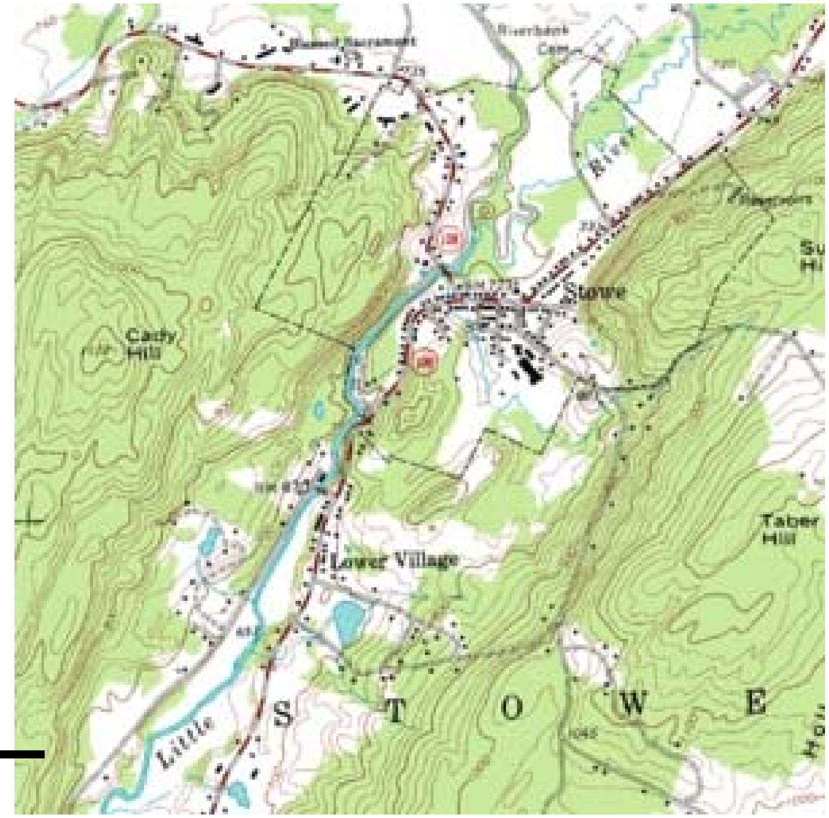
Interpretation of topographic maps
&
Effects of topography on outcrops

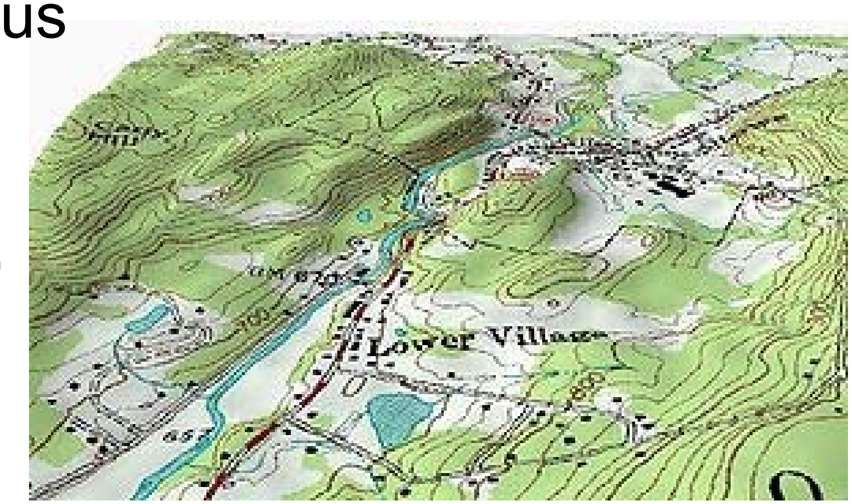
Topographic map

- A **topographic map** is a type of map characterized by large-scale detail and quantitative representation of relief, usually using contour lines.
- It is a detailed and accurate graphic representation of cultural and natural features on the ground.



Contour lines: Line representing locus of points in the map of equal value for a specific parameter.

Specific parameter: Heat, Pressure, rainfall, elevation, thickness etc.



In topographic maps: It is elevation with respect to mean sea level.

- Topographic maps are also called contour maps.

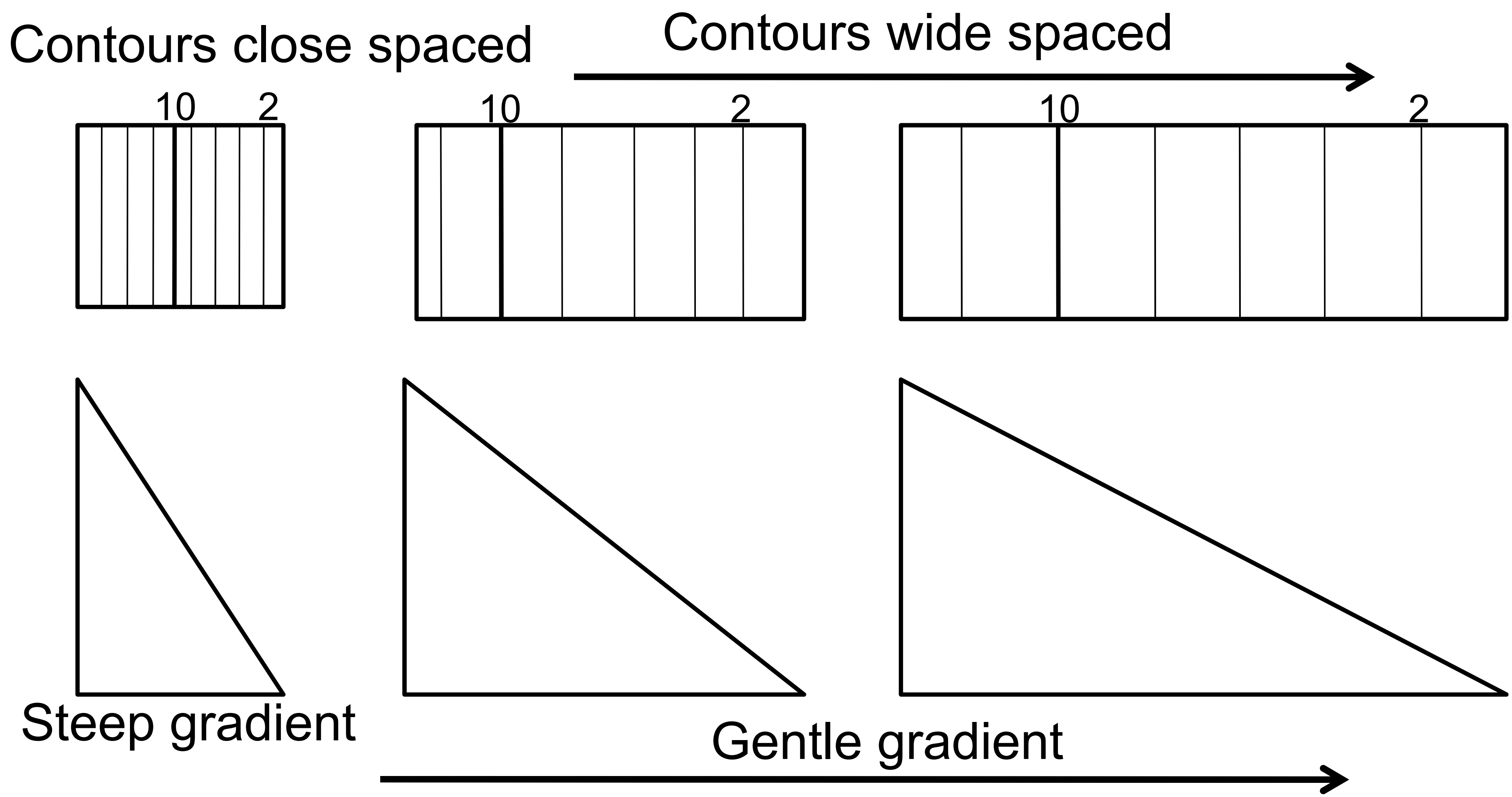
- A contour line on a topographic map can be envisioned as intersection of a horizontal plane with the ground surface.

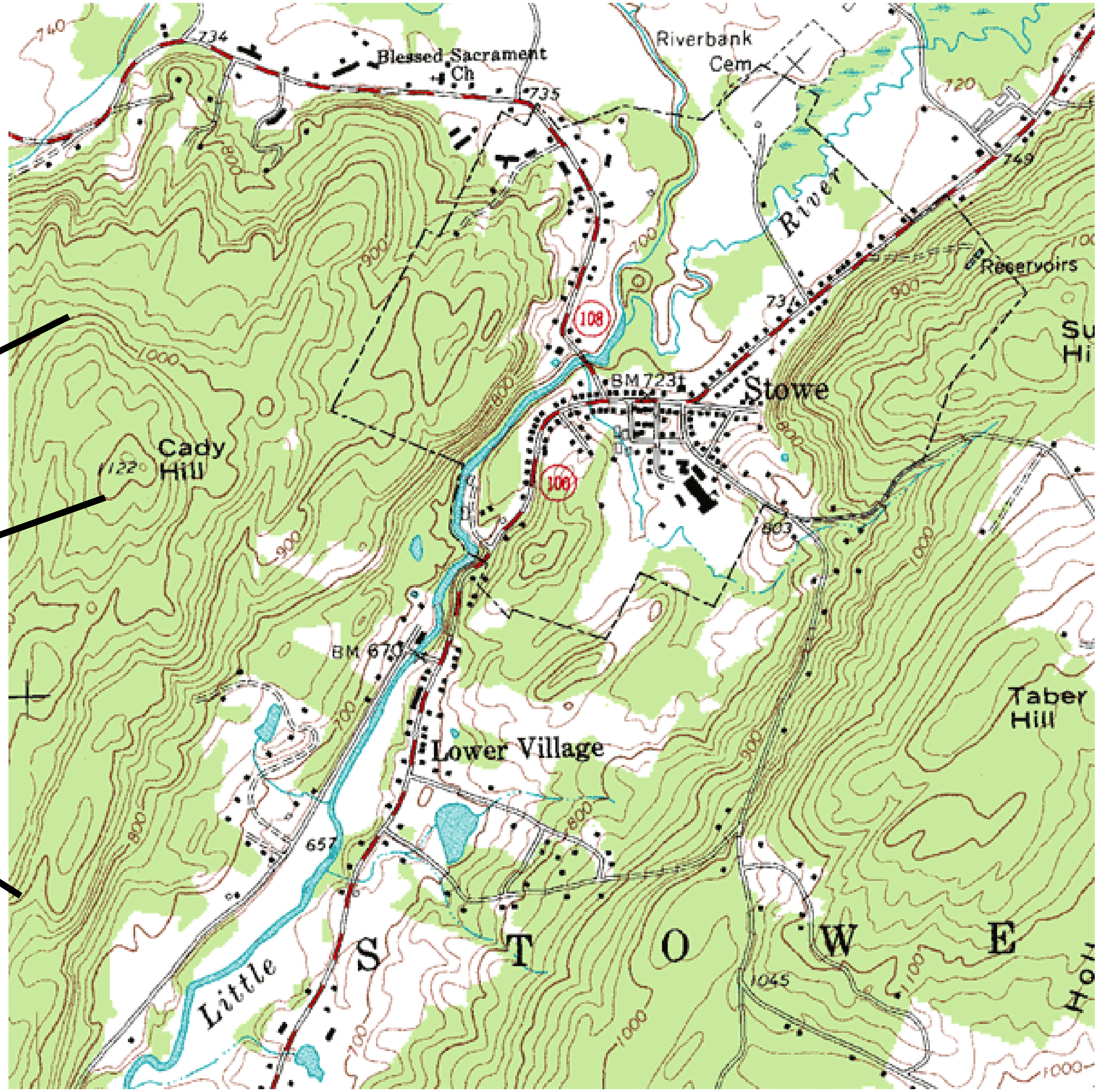
Elements of contour maps:

(i) Contour intervals & (ii) Gradients

Contour intervals: is the difference in the values of a parameter represented by adjacent contour lines.
In Topographic maps : its difference in elevation between adjacent contour lines.

Gradient : i.e. rate of change is directly proportional to spacing of contours.
Closely spaced contours: Steep gradient
Widely spaced contours: Gentle gradient


Contours close spaced
Contours wide spaced
10
2
10
2
10
2
Steep gradient
Gentle gradient



Contour lines

Index Contour
(Every 5th Contour)

Uses

Topographic maps have multiple uses in the present day: any type of

(i) geographic planning

(ii) large-scale architecture;

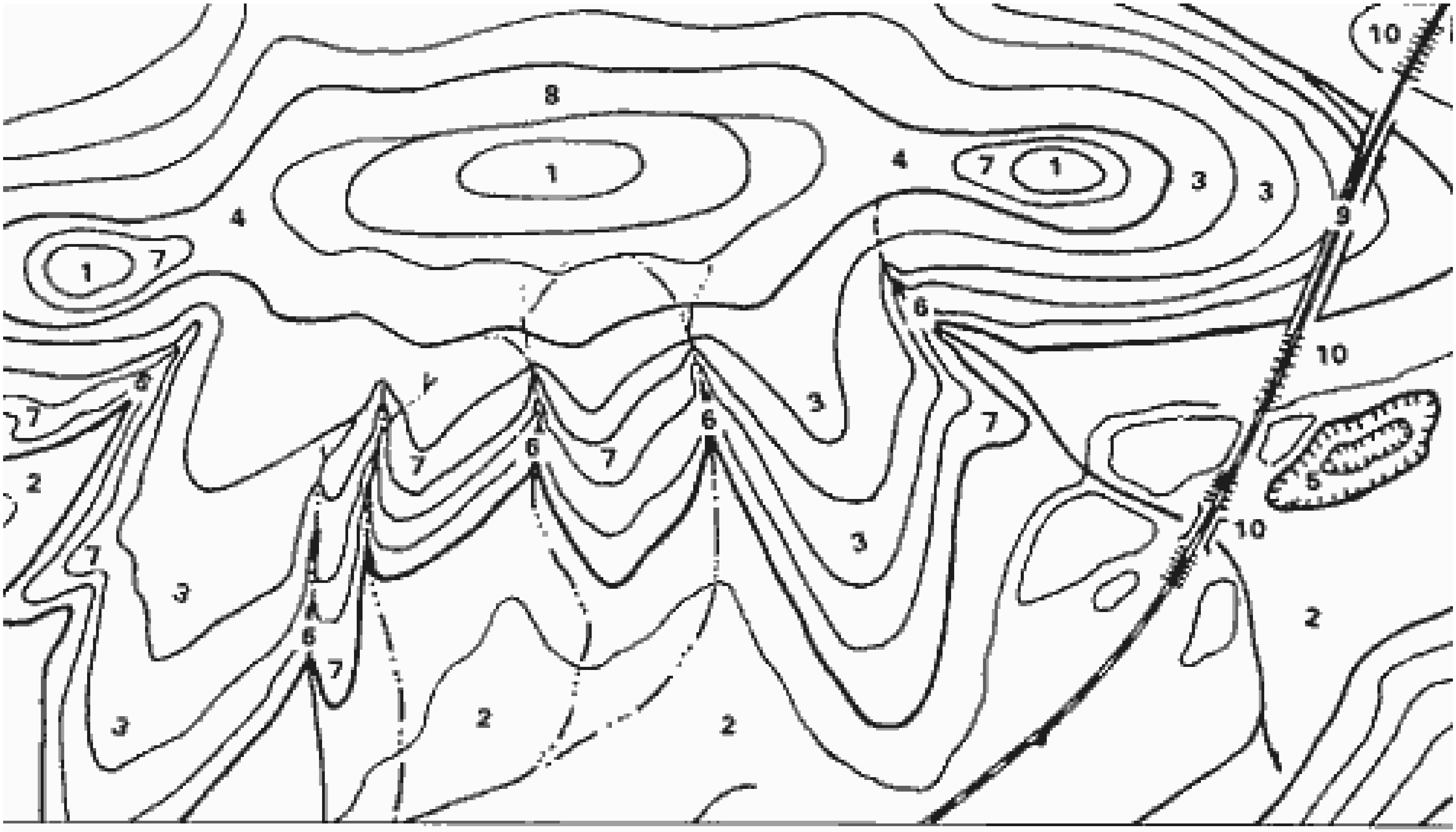
(iii) earth sciences and many other geographic disciplines;

(iv) mining and other earth-based endeavours; and

(v) recreational uses such as hiking or, in particular, orienteering, which uses highly detailed maps in its standard requirements.

In India: The Survey of India is responsible for all topographic control, surveys and mapping of India.

Study the given topographic map



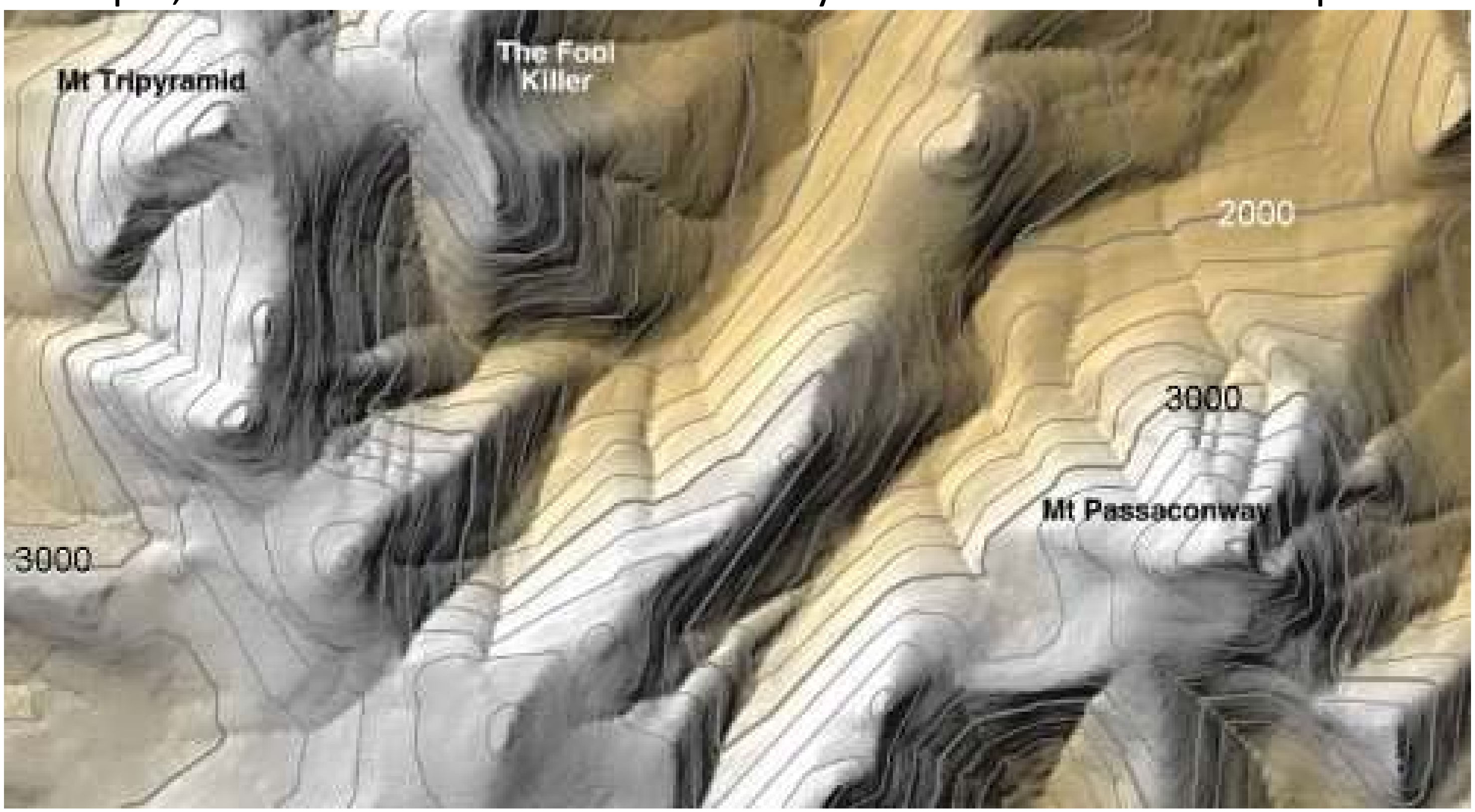
1. HILL
2. VALLEY
3. RIDGE
4. SADDLE
5. DEPRESSION
6. DRAW
7. SPUR
8. CLIFF
9. CUT
10. FILL

A Topographic Map includes *contour lines* drawn to represent changes in elevation. When you follow a path on a topographic map that crosses these contour lines, you will be either climbing or descending. A path running parallel to contour lines is relatively flat. When reading a topographic map, you need to visualize in your mind's eye a 3-dimensional view of what the symbols and contour lines are representing.

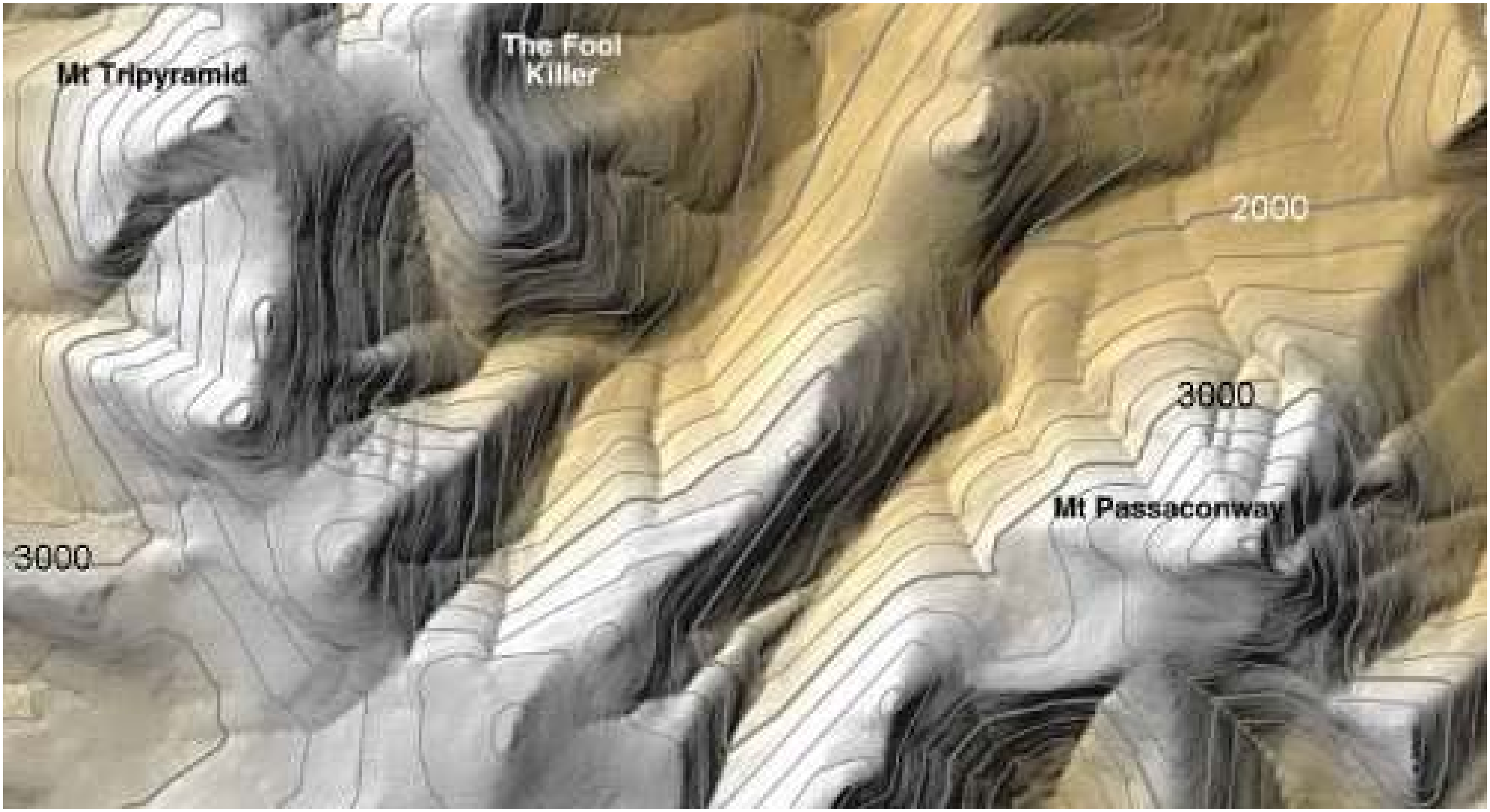
The most important thing to remember is that CLOSE contour lines mean STEEP terrain and OPEN contour lines mean FLAT terrain.

Shaded relief added to a topographic map makes it more realistic and helps visualize the real landscape.

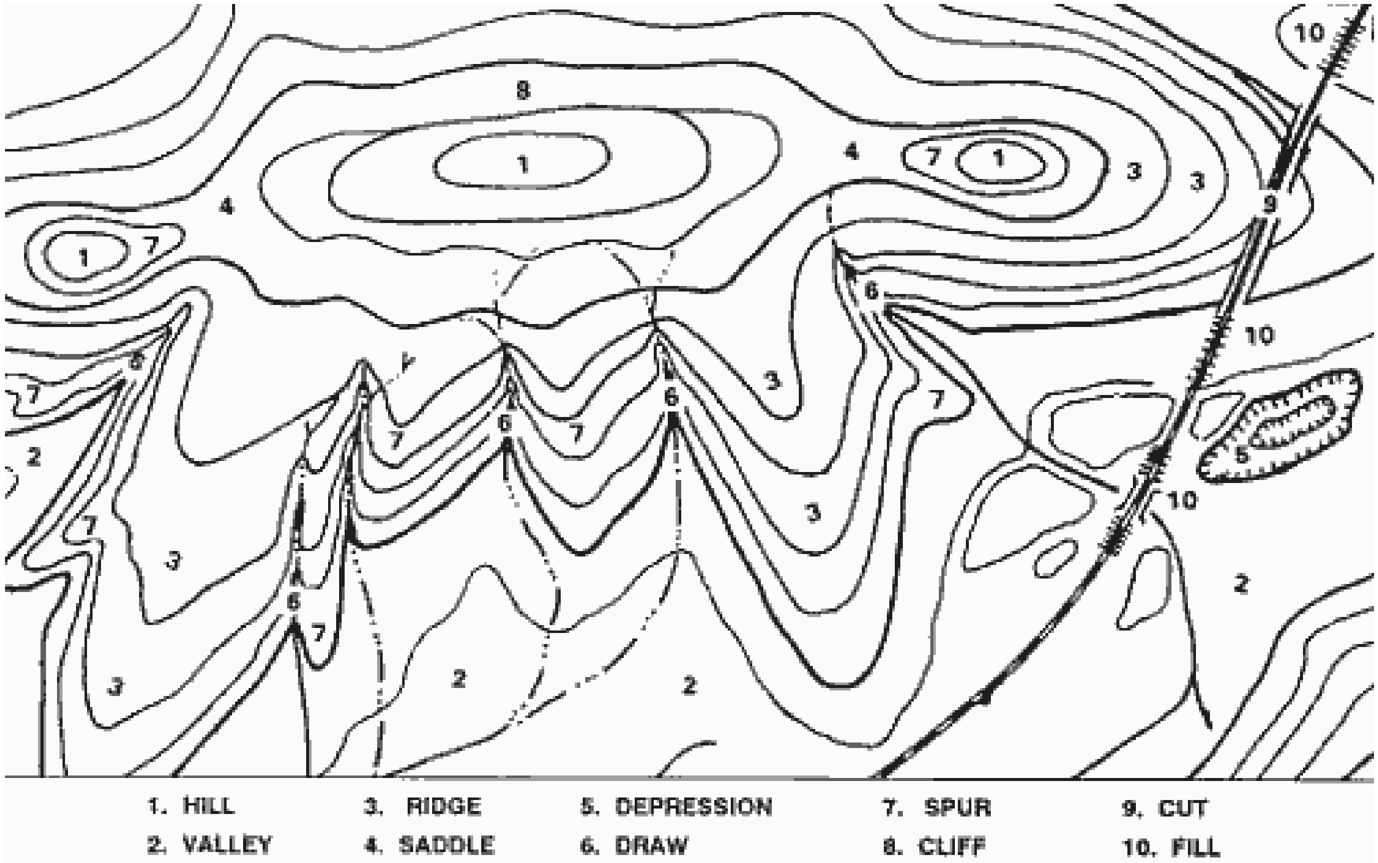
For example, see how the mountains and canyons stand out on this map:



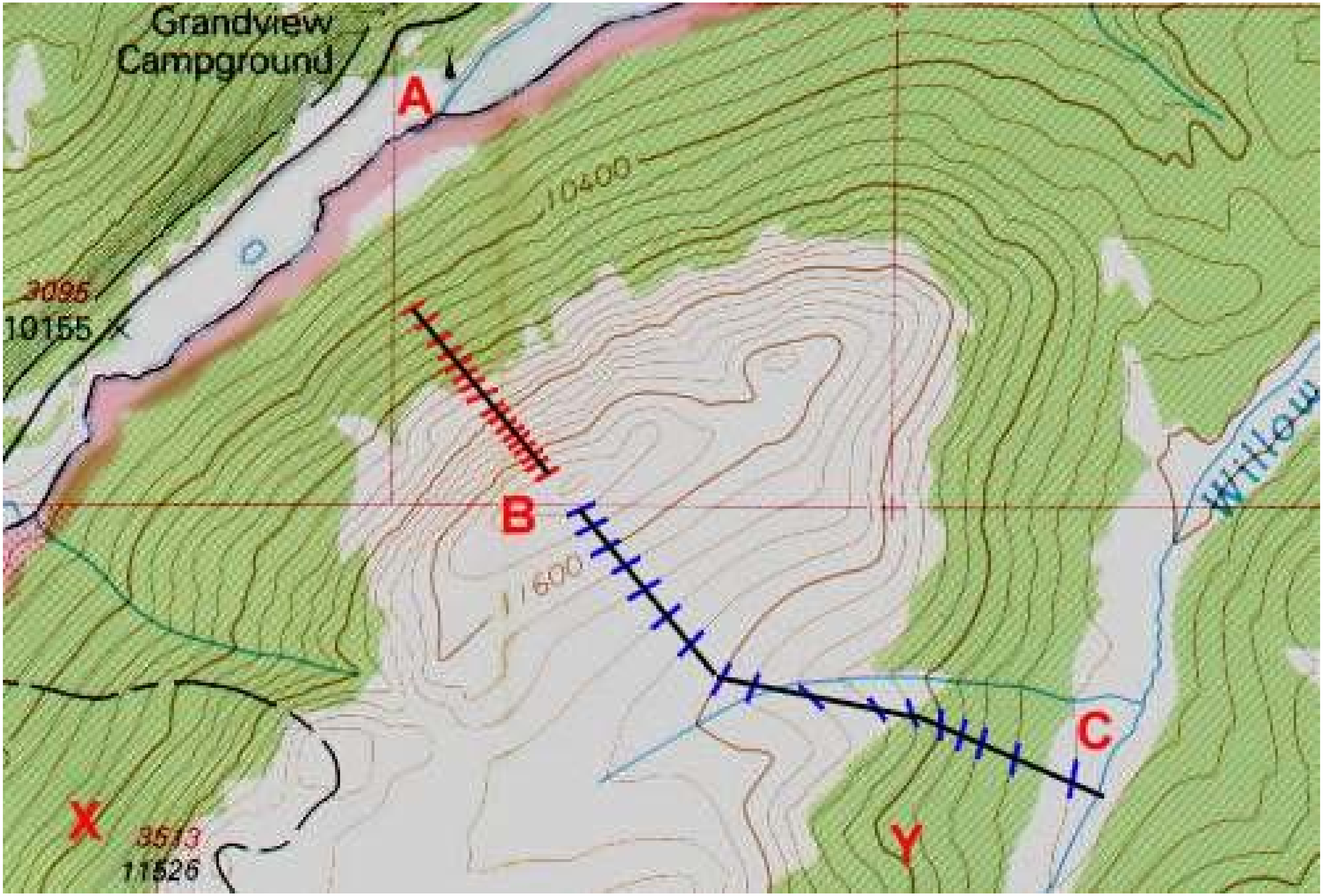
For example, see how the mountains and canyons stand out on this map:



- This example of a very simple topographic map shows many common features. Keep your eyes open to see these features on other maps and you will start to understand how a topo map works.



- Even without elevation numbers, clues that #1 is a hill include streams converging away from the hilltop, contour lines pointing sharply towards the hilltop (indicating draws), contour lines pointing widely away from the hilltop (indicating rounded ridges).



Grandview
Campground
A
10400
3095
10155
B
11600
Willow
C
X
3513
11526
Y

Using contour lines, you can tell a lot about the terrain, including steepness, ruggedness, and ground cover. On the image above, look at point **A**. There are no contour lines around this location so it is relatively flat here and a good place for a campground by the lake. You can tell from the elevation listed at marker 3095 that the campground is at 10155 feet.

You can also tell the elevation change between each contour line by looking at the Index lines. Notice that the Index line near point **B** is labeled 11600 feet and the one due north of it is labeled 10400 feet - that is a difference of 1200 feet. Between these two Index lines are two more Index lines so each index line represents a change in 400 feet of elevation - 10400, 10800, 11200, and 11600.

Count the lines between two index lines and you should see there are 4 lines which cause the 400 feet between the two index lines to be divided into 5 intervals, each one being 80 feet in elevation. So, now we know that *on this map* every contour line represents 80 feet of elevation change.

If you follow a single contour line, your elevation remains constant. For example, starting at point **X** and following the Index line to the NorthEast, around, and down South to point **Y**, you would stay at about 10,800 feet.

When you cross contour lines, you are either hiking up or down. Look at the two routes to get to the peak at point **B** - the red route and the blue route. Each path reaches the top, but the blue route is three times as long as the red route. That means it covers more distance to gain the same elevation so it is a more gradual slope - and probably an easier hike. Going up the red route may require a lot of scrambling and hard work.

General constraints on contour maps:

- There are several general rules that constrain the construction of contour maps:

1. The contour interval on a map is constant.

The difference in the value of a parameter (i.e. elevation in topographic maps) represented by two adjacent contour lines should be the same everywhere on the map.

If not: difficult to calculate gradients.

Exception: Domains of both very steep and very shallow gradients.

(i) Steep gradients: Large contour interval is selected to accommodate domains of the map in which there are steep gradients.

(ii) Shallow gradients: Small contour intervals can be added to provide greater resolution of features.

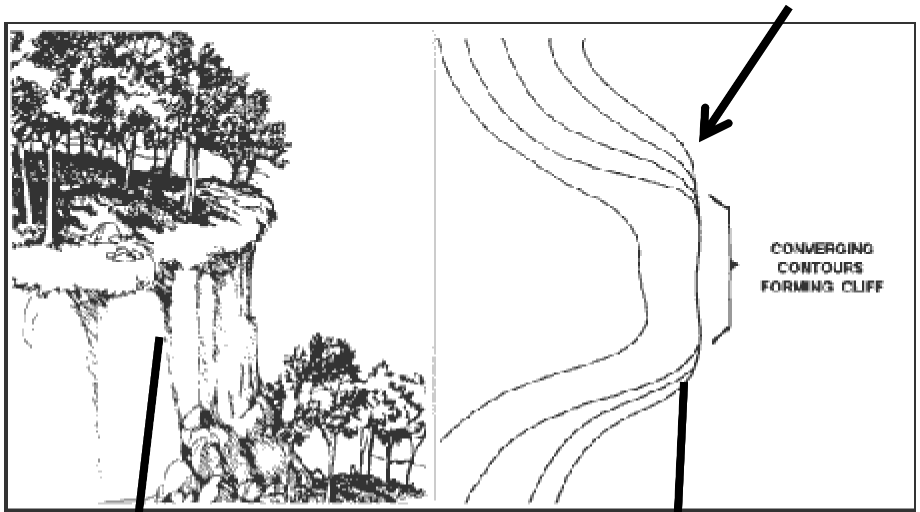
2. Contour Lines generally should not merge or cross.

Contour lines connect areas of equal elevation. Therefore, if a contour line were to cross another line, it would mean that the point where they cross has two different elevations. This cannot occur. So the map may be wrong.

Exception: Two acceptable situations

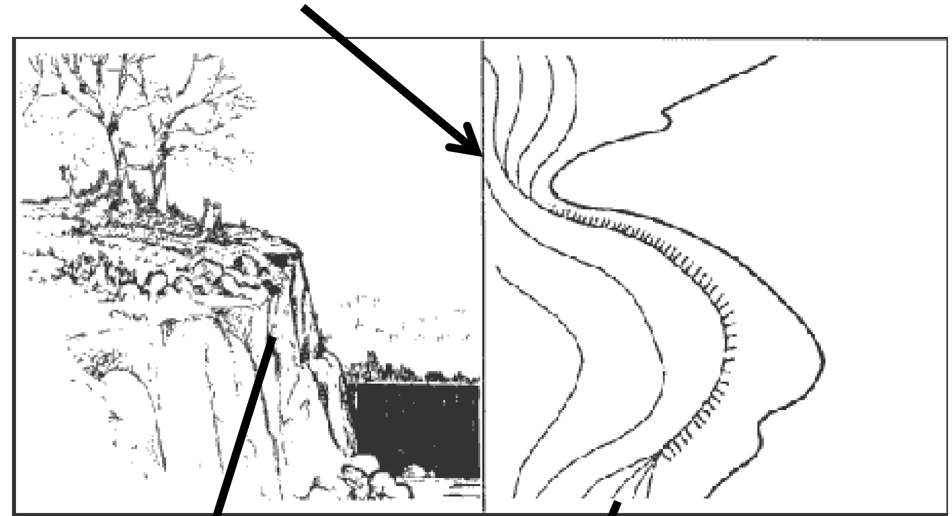
-(i) When there is an overhanging cliff face. In this case, the two lines can cross, and the line of lower elevation is marked as dashes.

-(ii) If there is a vertical cliff (i.e. if gradient is infinite), contour lines merge on a map



vertical cliff

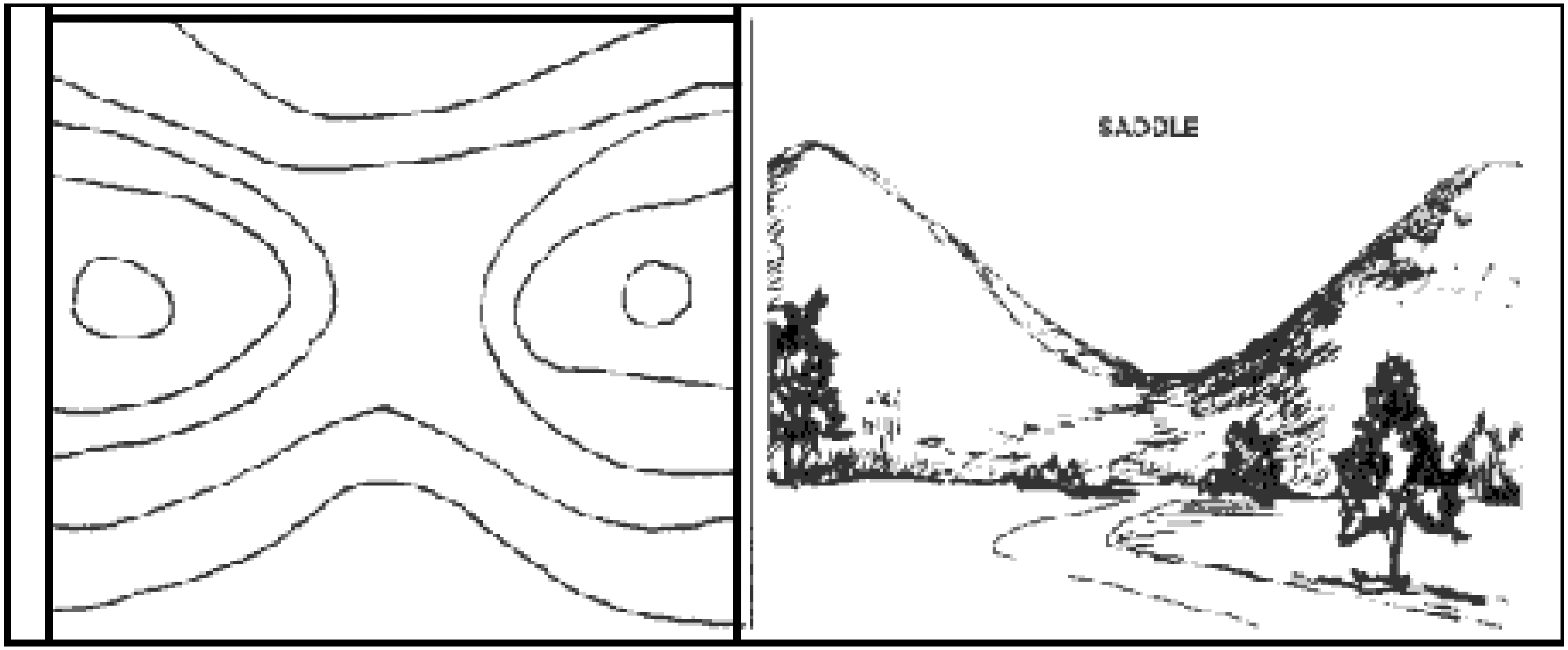
contour lines merge



vertical cliff

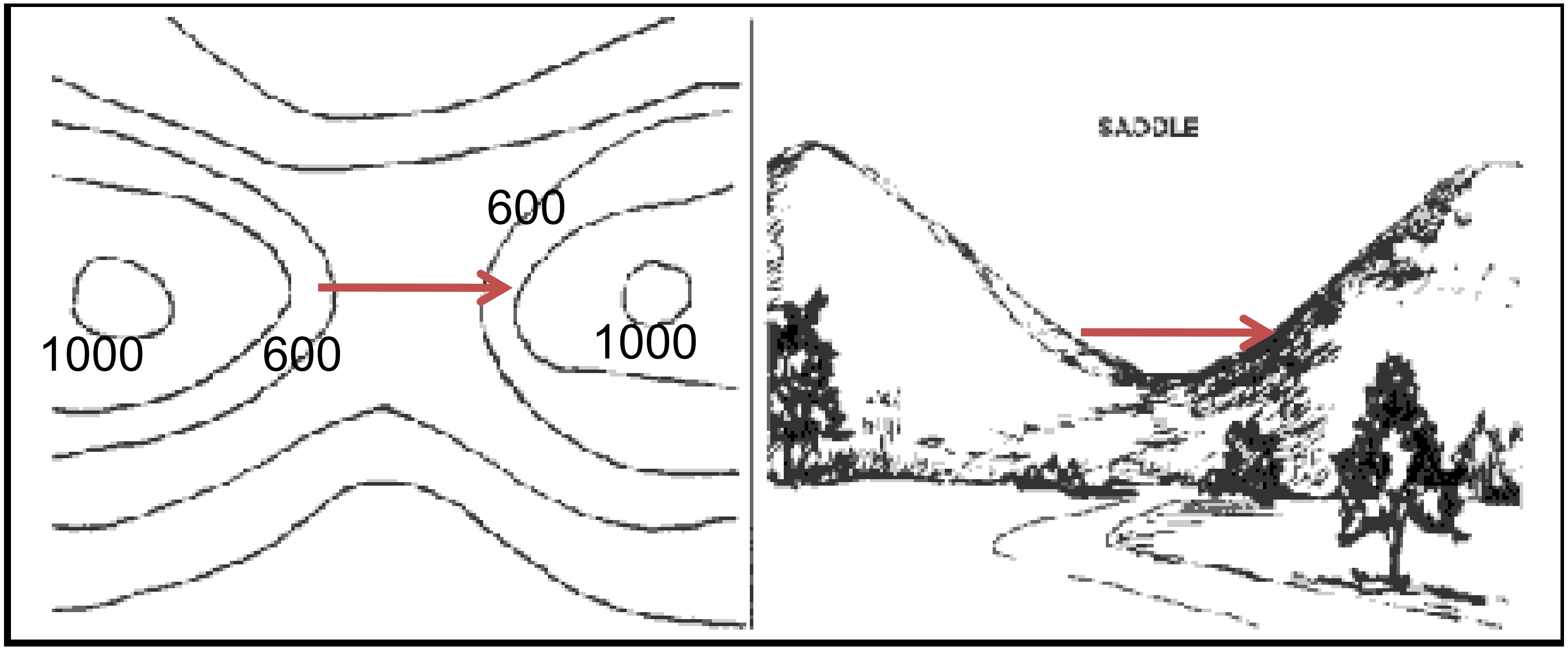
contour lines merge

3. Contour lines either close within the area of the map or are truncated by the edge of the map or by a structure.



4. Contour lines are repeated to indicate reversals in gradient direction.

If along a traverse line, direction of a gradient reverses, the lowest (or highest) contour crossed before the reversal must be repeated after reversals.



5. A reference frame for a contour is defined by specifying a datum plane.

A datum plane, also called reference plane, for a contour map is an imaginary surface on which the parameter described by the map has a value of “0”.

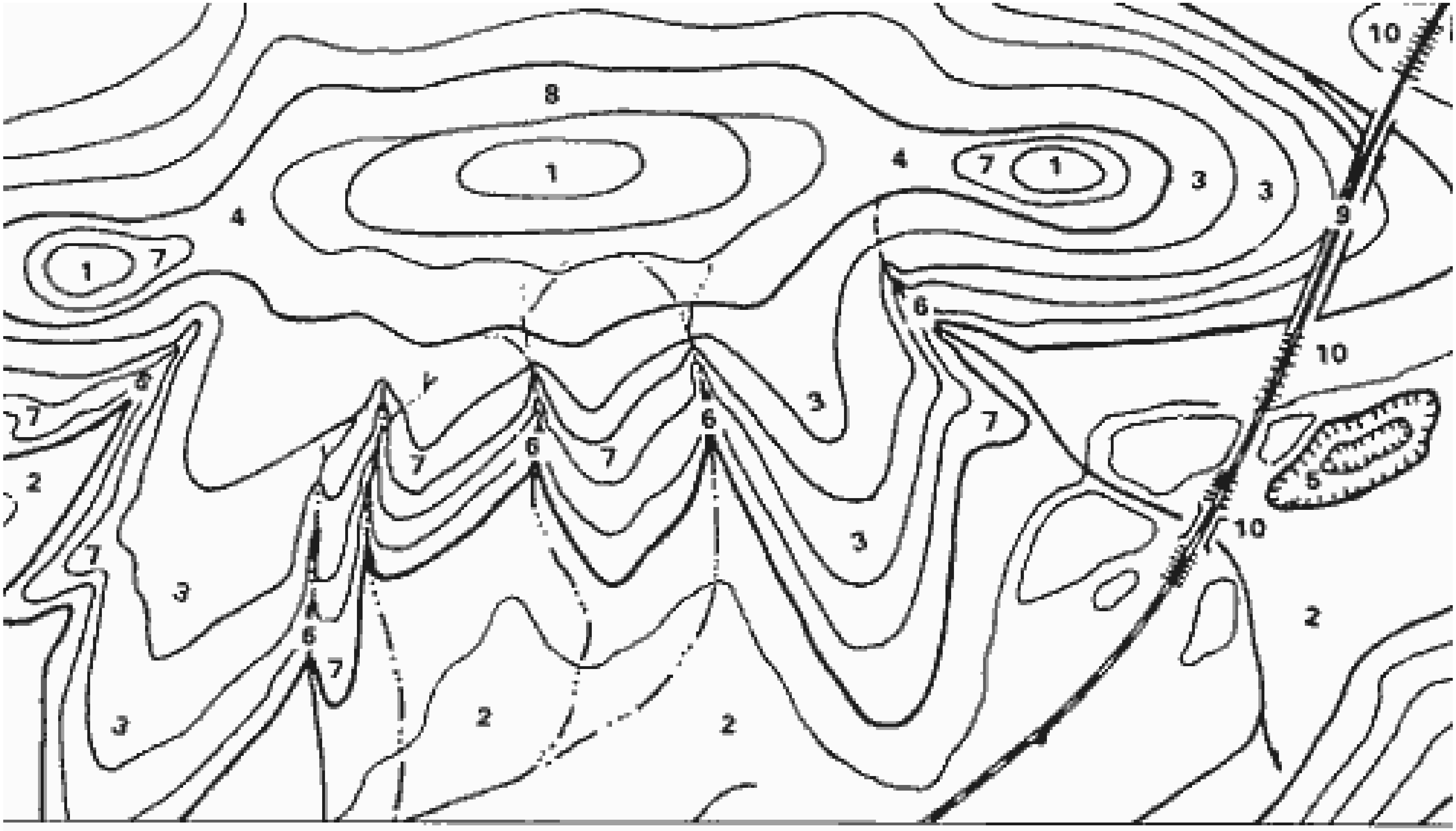
eg. on a topographic map the datum plane is mean sea level.

Topographic (i.e terrain) features

Major
Hill, Saddle, Valley, Ridge, Depression

Minor
Draw, Spur, Cliff

Supplementary
Cut, Fill



1. HILL
2. VALLEY
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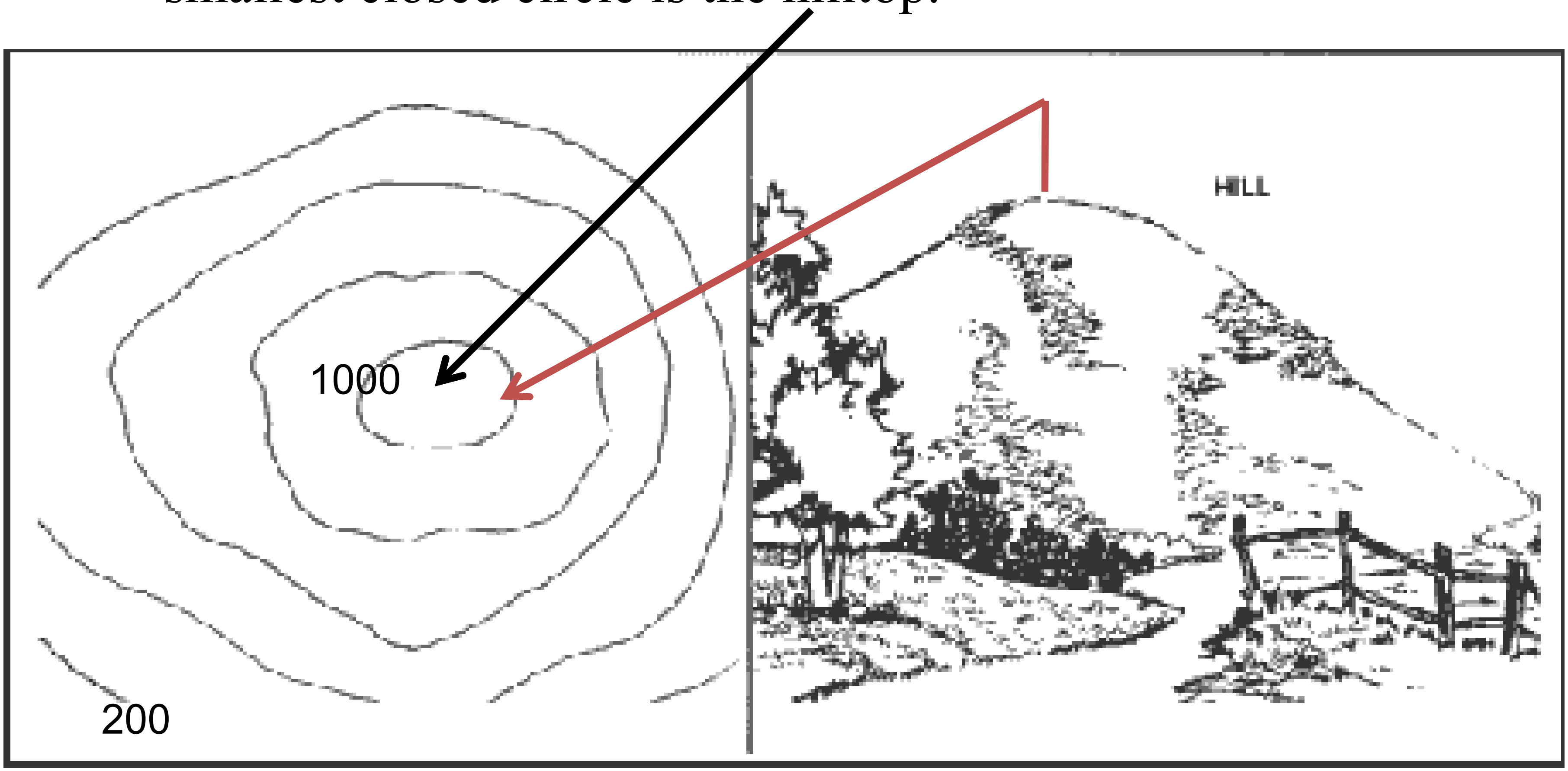
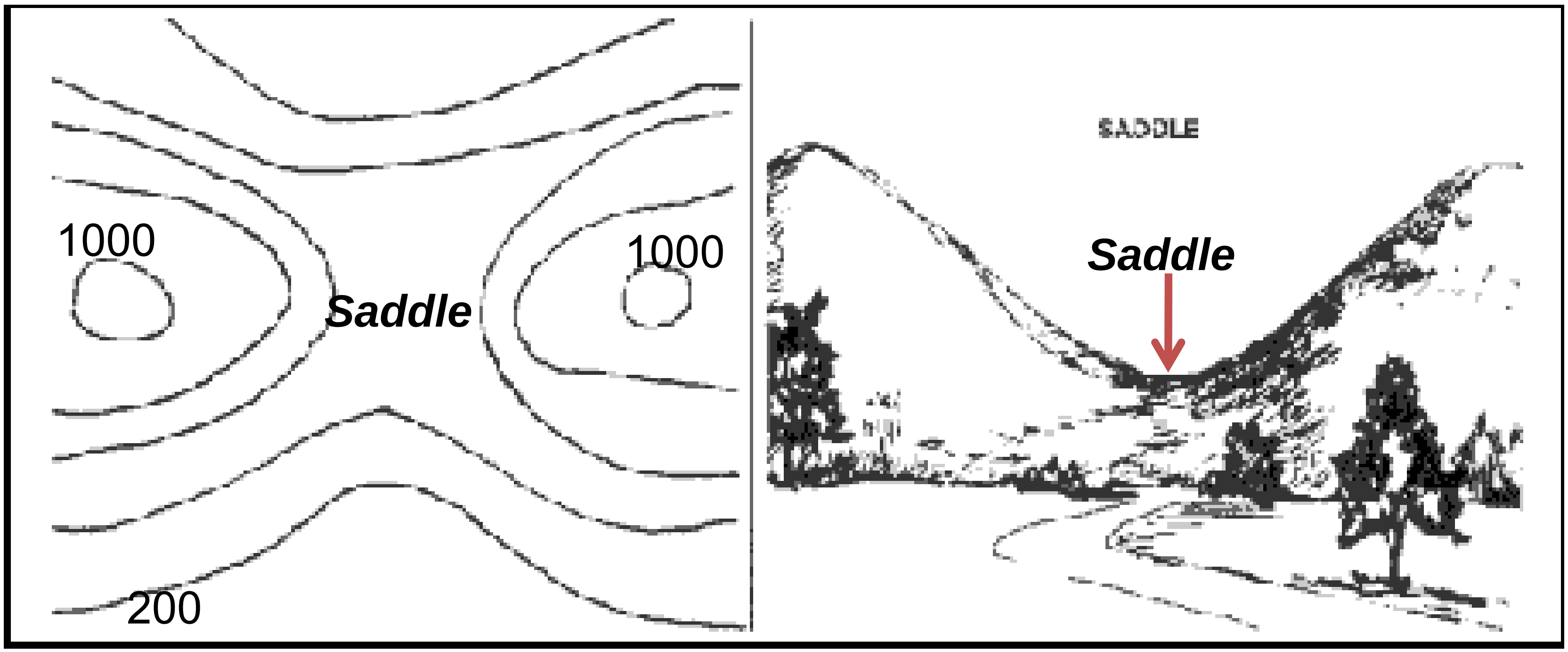
<u>Major topographic (i.e terrain) features</u>

1. <u>Hill</u>. A hill is an area of high ground. From a hilltop, the ground slopes down in all directions. A hill is shown on a map by contour lines forming concentric circles. The inside of the smallest closed circle is the hilltop.



(2) ***Saddle***. A saddle is a dip or low point between two areas of higher ground.

If you are in a saddle, there is high ground in two opposite directions and lower ground in the other two directions.



Valley.

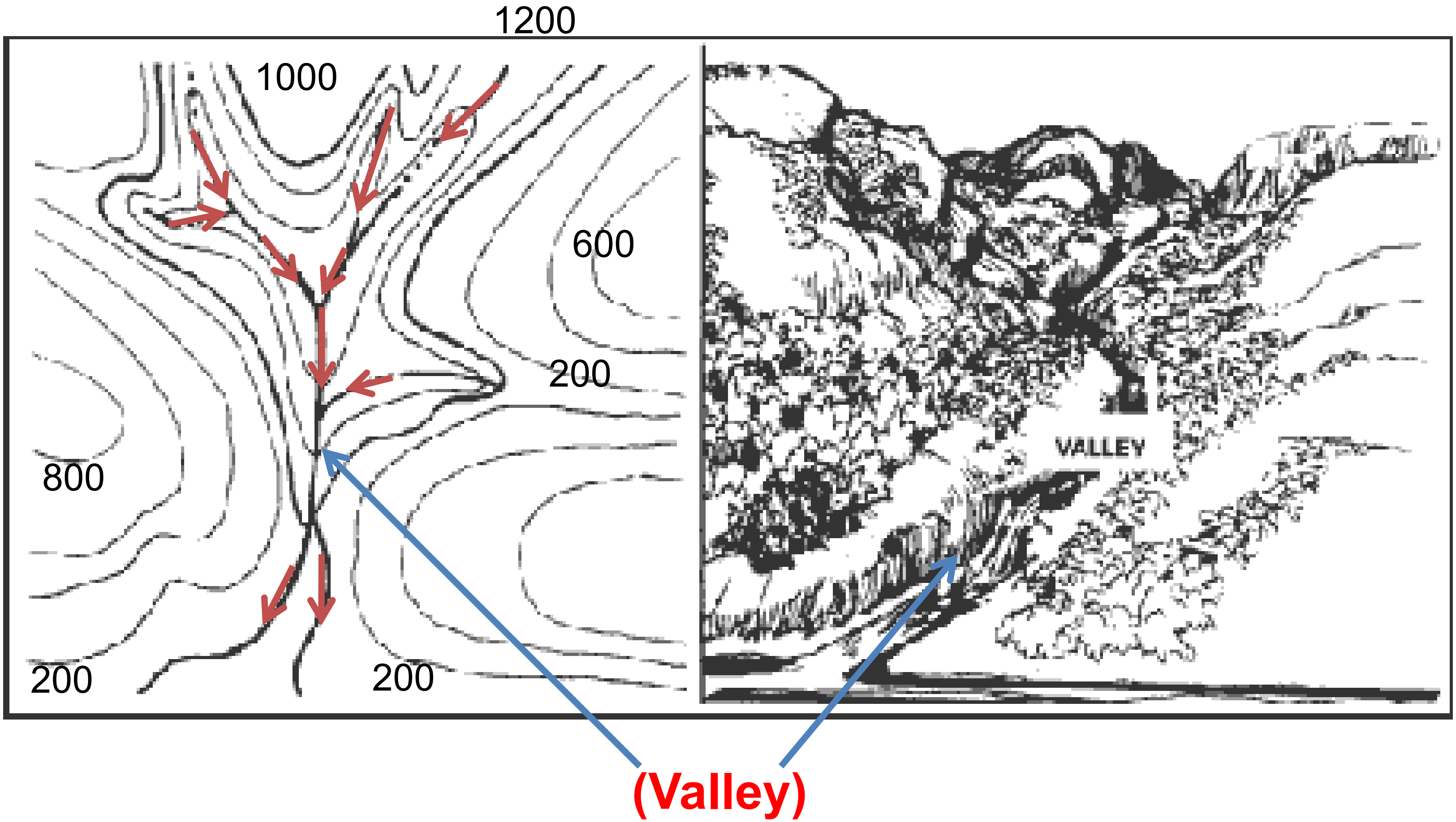
- A valley is a stretched-out groove in the land, usually formed by streams or rivers.

- A valley begins with high ground on three sides, and usually has a course of running water through it.

- If standing in a valley, three directions offer high ground, while the fourth direction offers low ground.

- Contour lines forming a valley are either U-shaped or V-shaped.

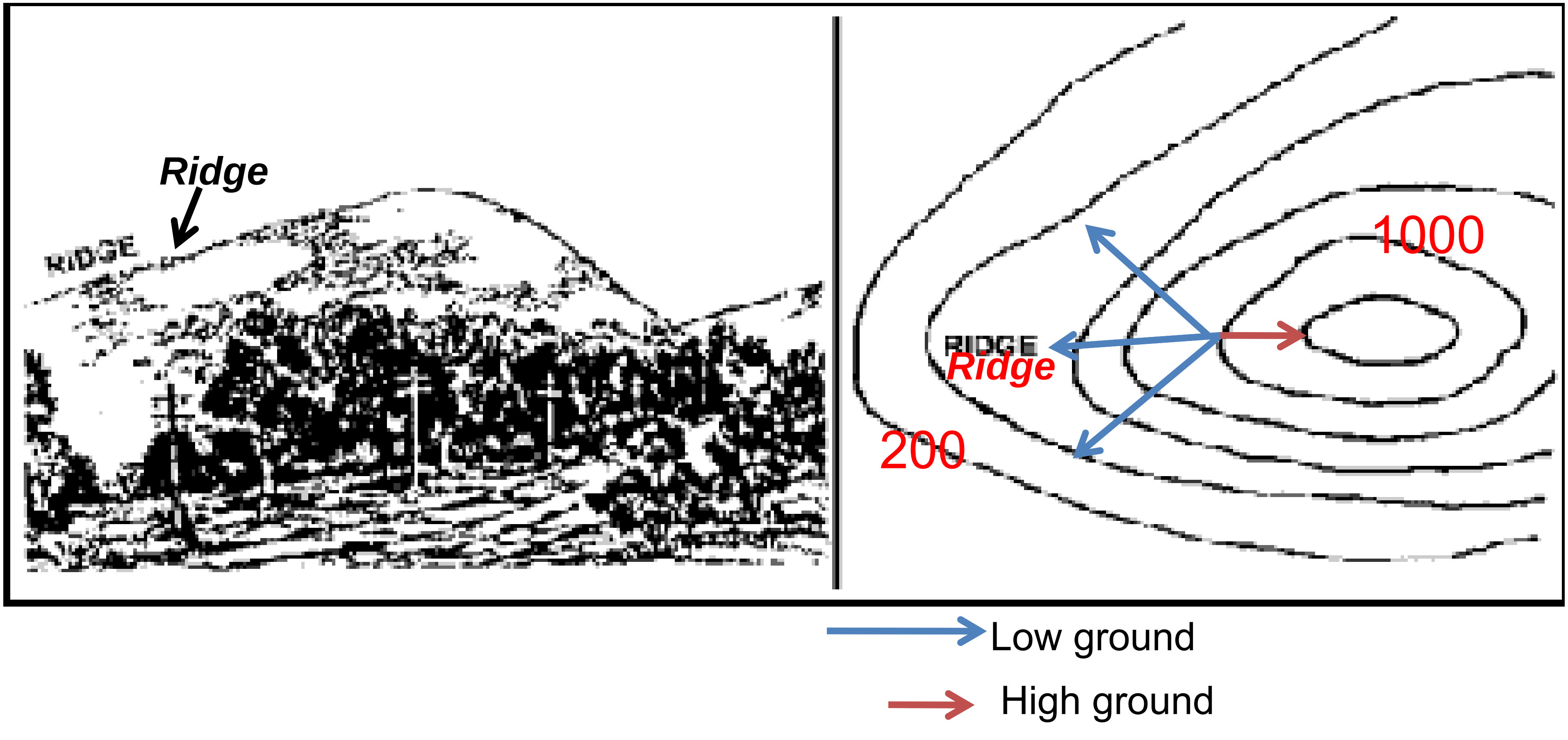
- To determine the direction water is flowing, look at the contour lines.
- The closed end of the contour line (U or V) always points upstream or towards high ground.



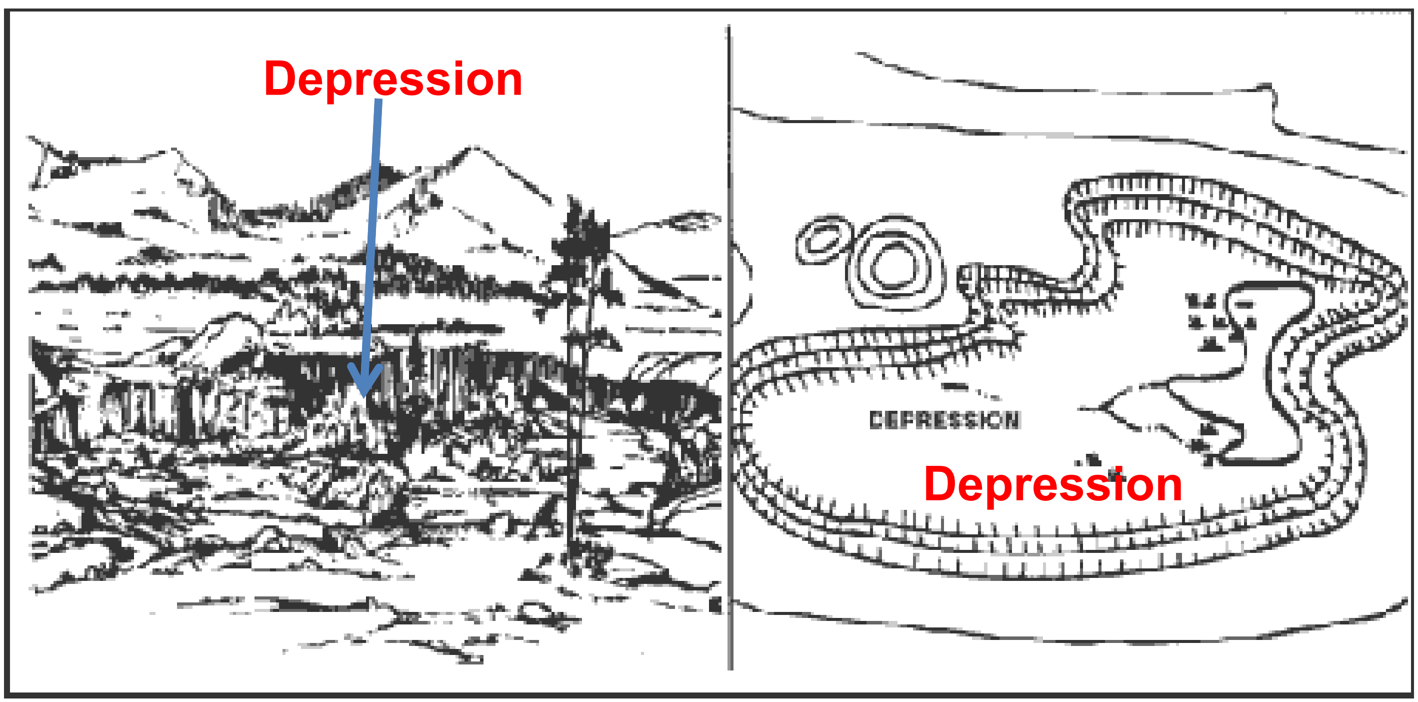
Direction of water flow in stream

-***Ridge***. A ridge is a sloping line of high ground.

- If you are standing on the centre-line of a ridge, you will normally have low ground in three directions and high ground in one direction with varying degrees of slope.
- If you cross a ridge at right angles, you will climb steeply to the crest and then descend steeply to the base.
- Contour lines forming a ridge tend to be U-shaped or V-shaped.
- The closed end of the contour line points away from high ground.

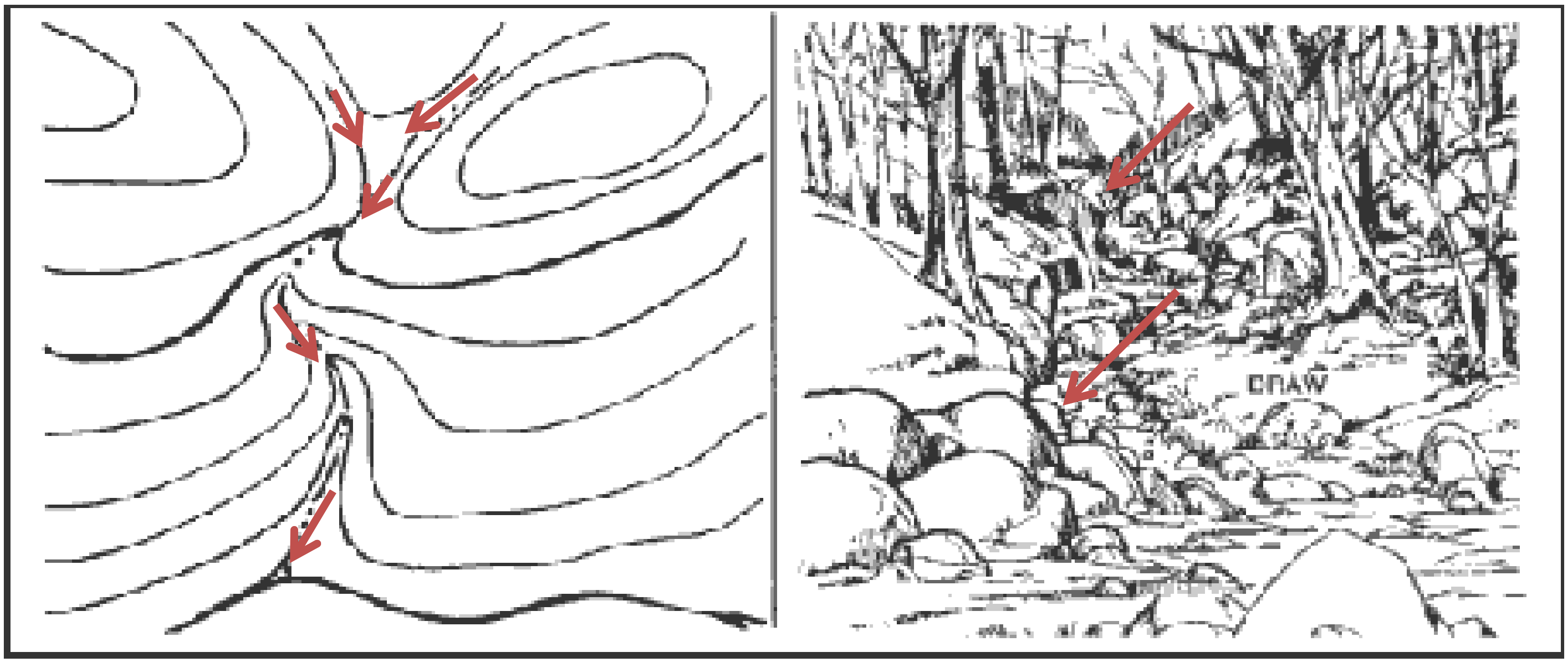


Depression: A depression is a low point in the ground or a sinkhole.It could be described as an area of low ground surrounded by higher ground in all directions or simply a hole in the ground. Usually only depressions that are equal to or greater than the contour interval will be shown. On maps, depressions are represented by closed contour lines that have tick marks pointing toward low ground.



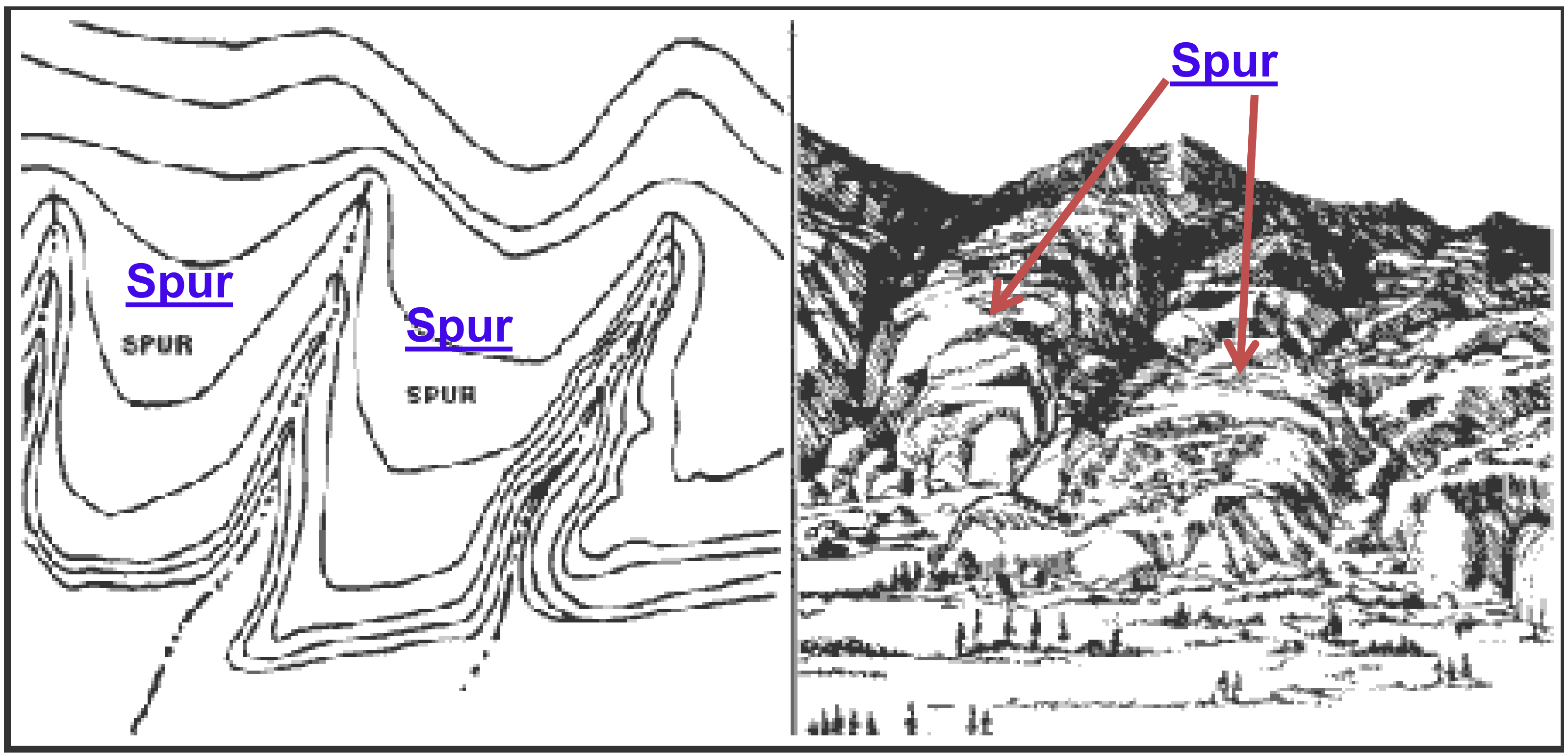
Minor Terrain Features.

<u>Draw</u>. A draw is a less developed stream course than a valley. If you are standing in a draw, the ground slopes upward in three directions and downward in the other direction. A draw could be considered as the initial formation of a valley. The contour lines depicting a draw are U-shaped or V-shaped, pointing toward high ground.

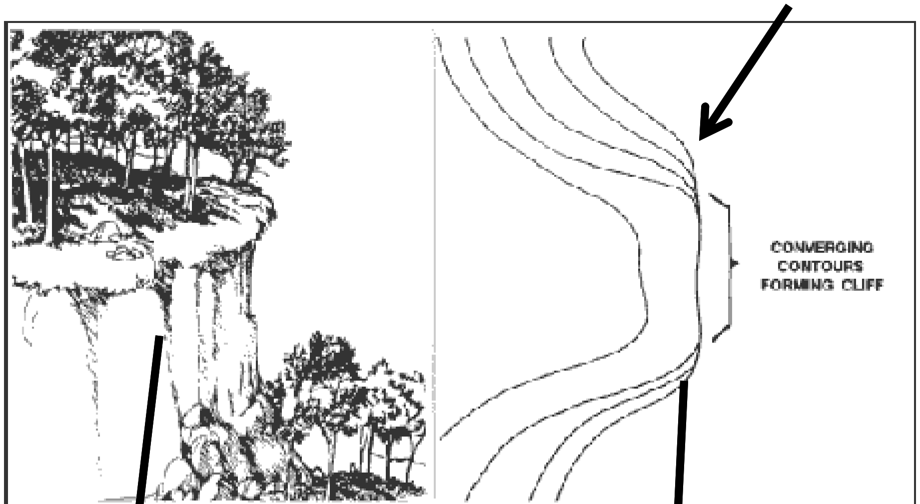


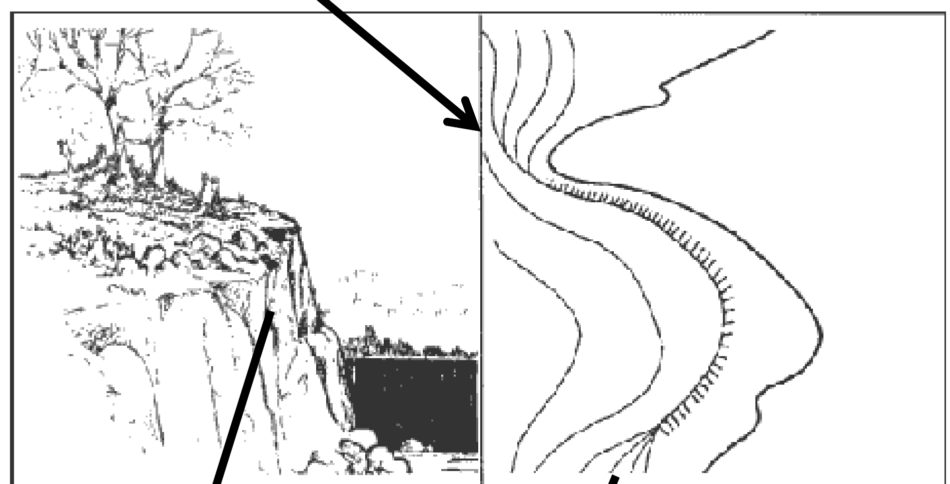
→ **Direction of water flow in stream**

Spur. A spur is a short, continuous sloping line of higher ground, normally jutting out from the side of a ridge. A spur is often formed by two rough parallel streams, which cut draws down the side of a ridge. The ground sloped down in three directions and up in one direction. Contour lines on a map depict a spur with the U or V pointing away from high ground.



Cliff. A cliff is a vertical or near vertical feature; it is an abrupt change of the land. When a slope is so steep that the contour lines converge into one "carrying" contour of contours, this last contour line has tick marks pointing toward low ground. Cliffs are also shown by contour lines very close together and, in some instances, touching each other.





vertical cliff

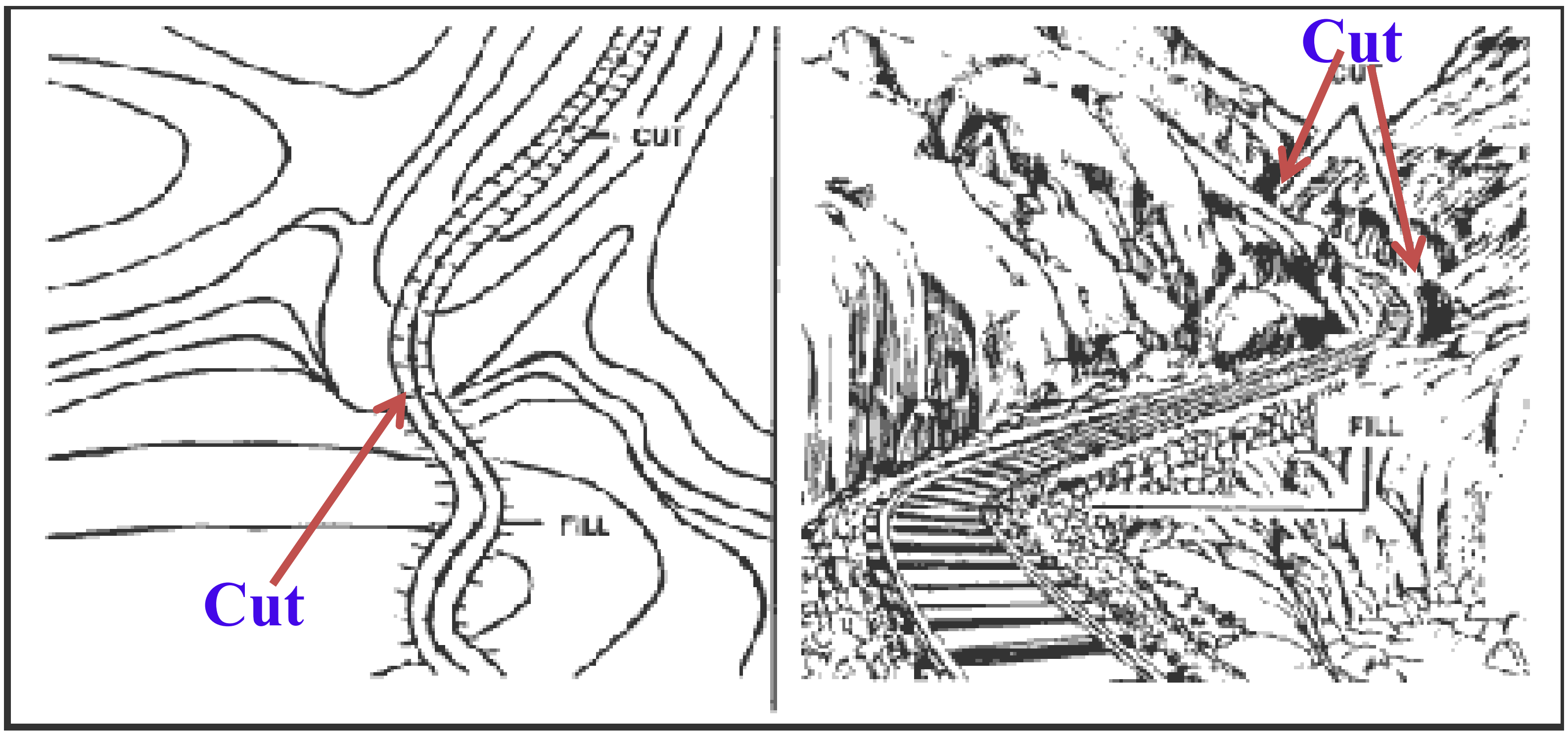
vertical cliff

contour lines merge

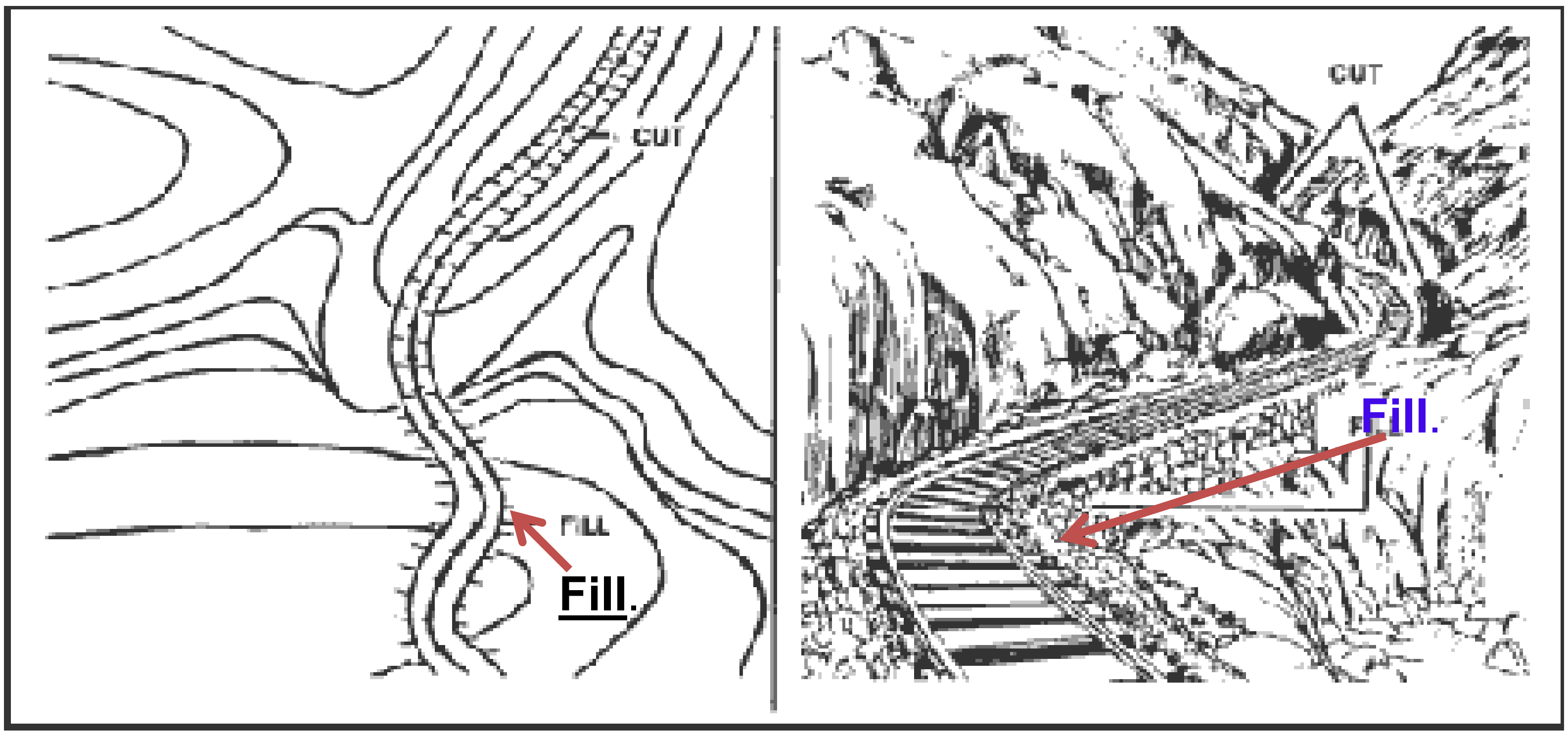
contour lines merge

Supplementary Terrain Features.

Cut: A cut is a man-made feature resulting from cutting through raised ground, usually to form a level bed for a road or railroad track. Cuts are shown on a map when they are at least 10 feet high, and they are drawn with a contour line along the cut line. This contour line extends the length of the cut and has tick marks that extend from the cut line to the roadbed, if the map scale permits this level of detail.



Fill. A fill is a man-made feature resulting from filling a low area, usually to form a level bed for a road or railroad track. Fills are shown on a map when they are at least 10 feet high, and they are drawn with a contour line along the fill line. This contour line extends the length of the filled area and has tick marks that point toward lower ground. If the map scale permits, the length of the fill tick marks are drawn to scale and extend from the base line



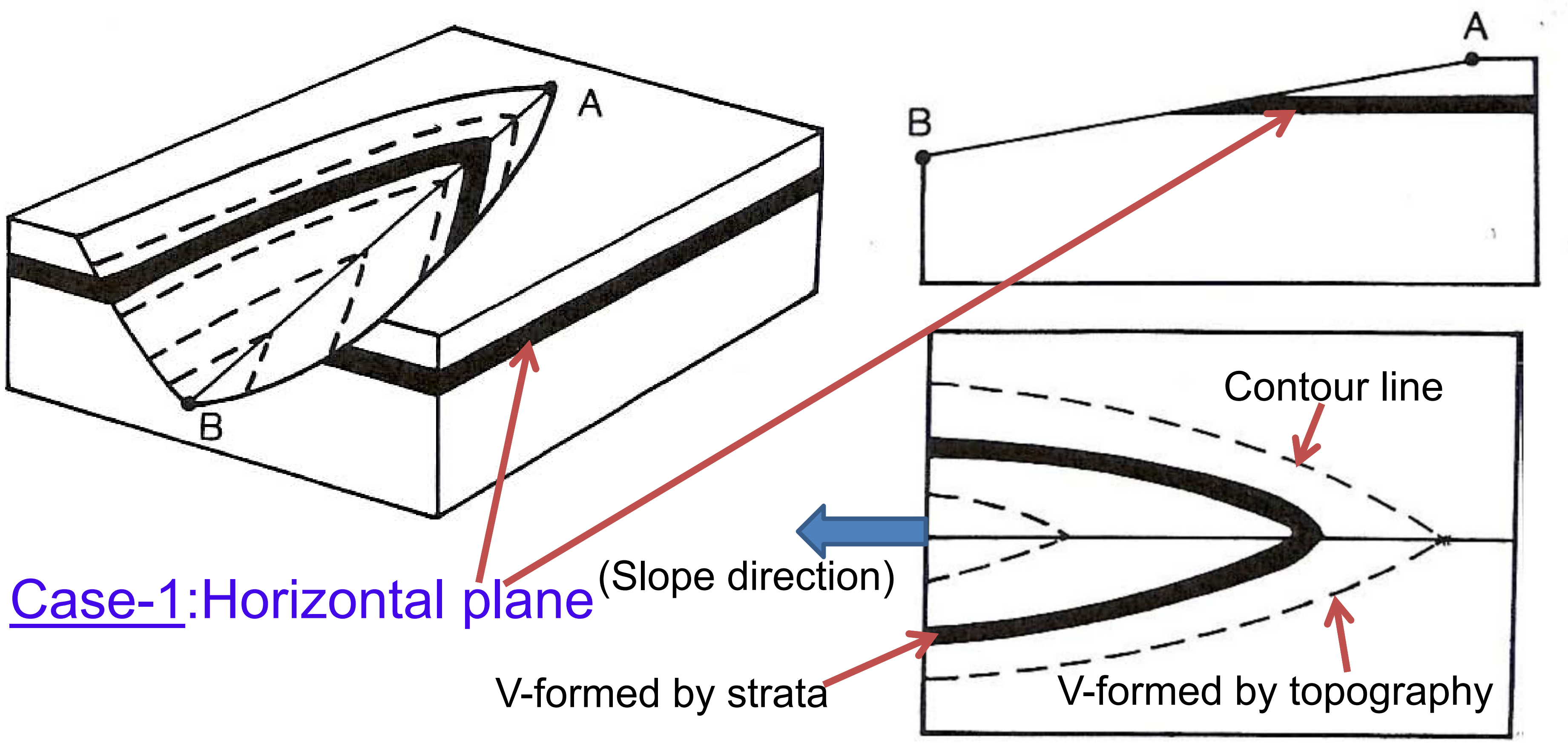
Interpretation of topographic maps
&
Effects of topography on outcrops

Effects of topography on outcrops

Case-1: Intersection of horizontal plane with topography

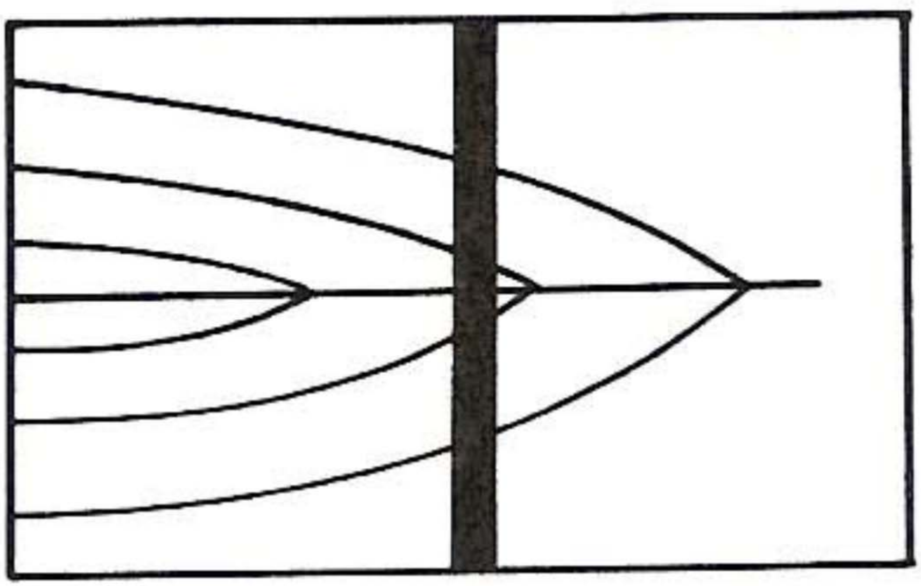
Figure 2-8. Intersection of a horizontal plane with topography. (a) Block diagram; (b) cross section along the axis of the valley; (c) map view showing that the outcrop trace is parallel to contour lines.

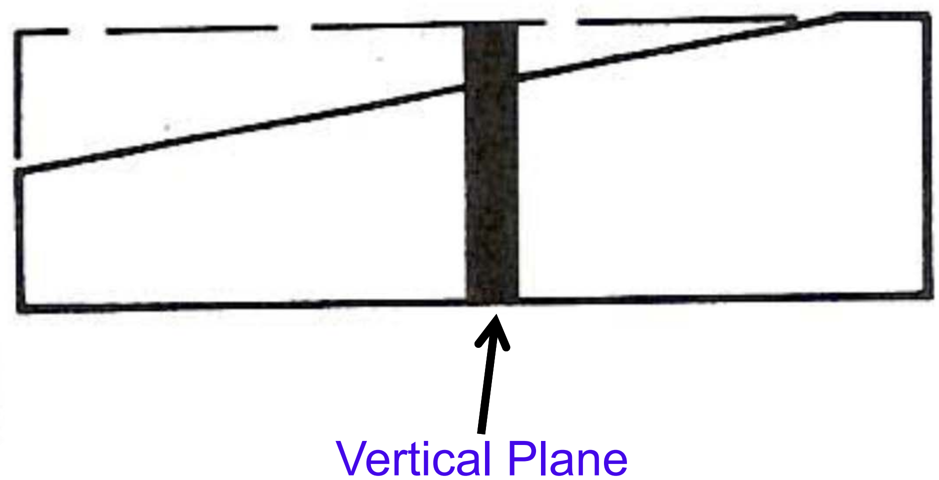
Topographic V and Bed V are parallel. Bed runs parallel to contour lines.



vertical plane;

Case 2: Vertical Plane

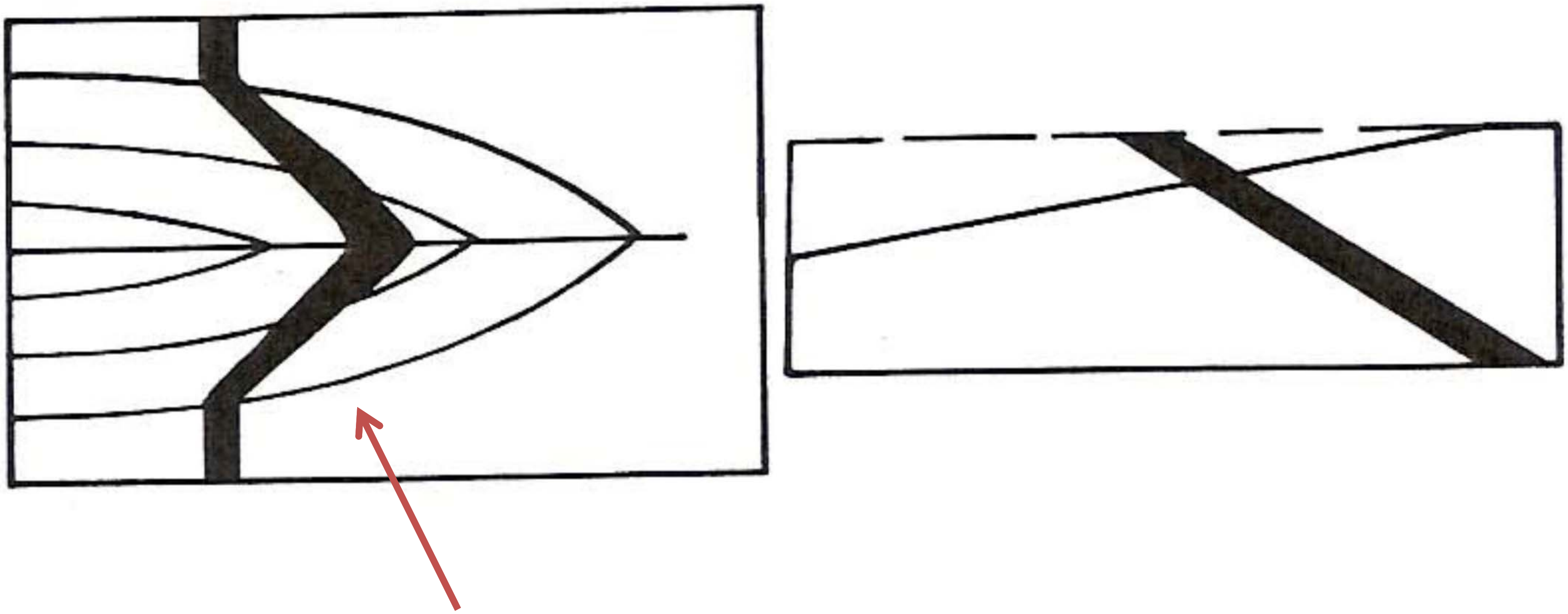




Bed/strata doesn't form any V. Bed runs line a straight line across both walls of the valley.

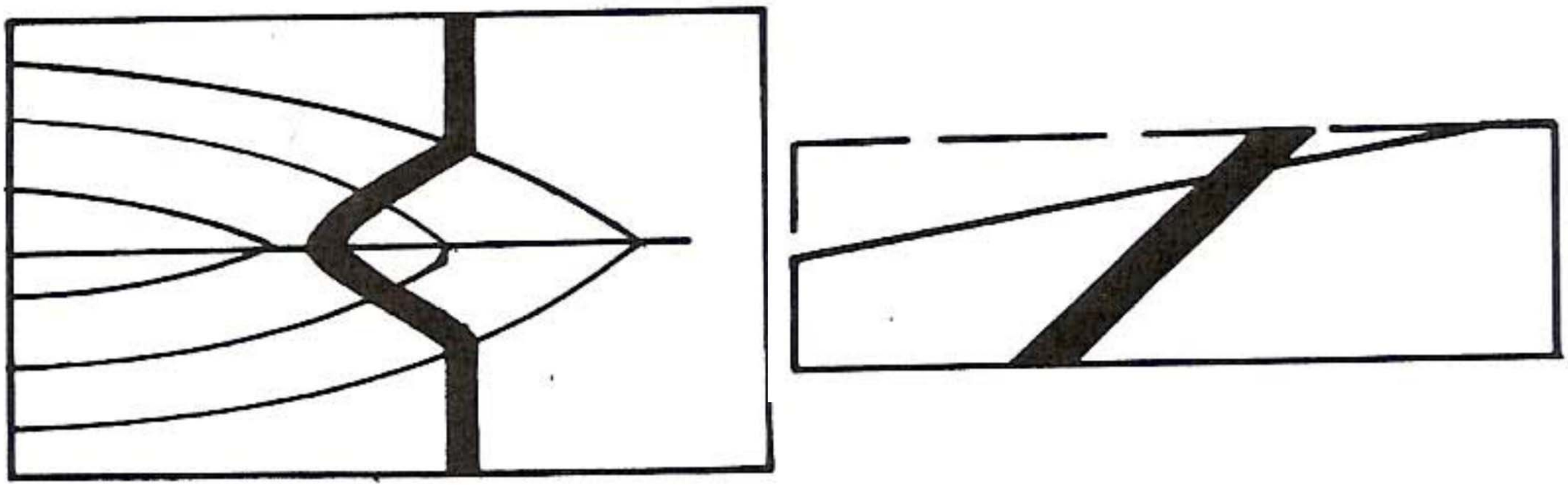
Plane dipping upstream;

Case 3: Plane dipping upstream



Topographic V and Bed V in same direction.
Wings of v's away each other.

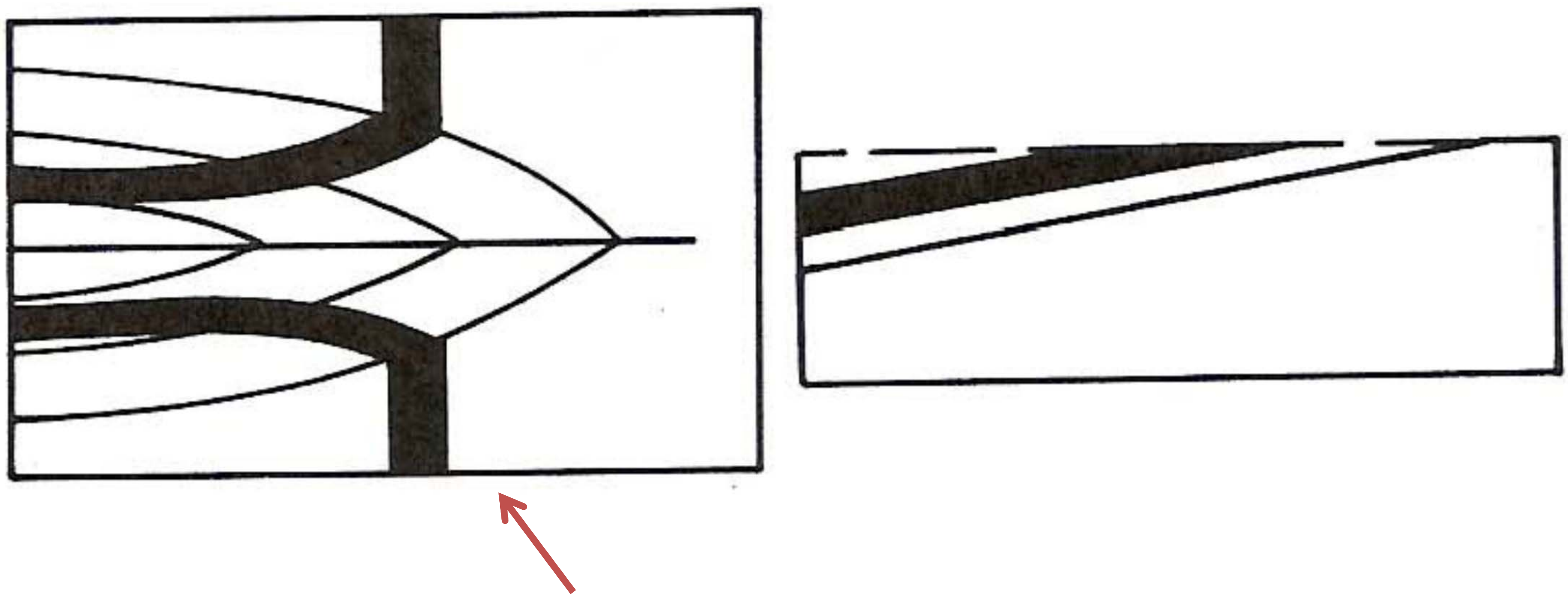
(c) plane dipping downstream at an angle greater than the stream gradient;



Topographic V and Bed V in opposite direction.

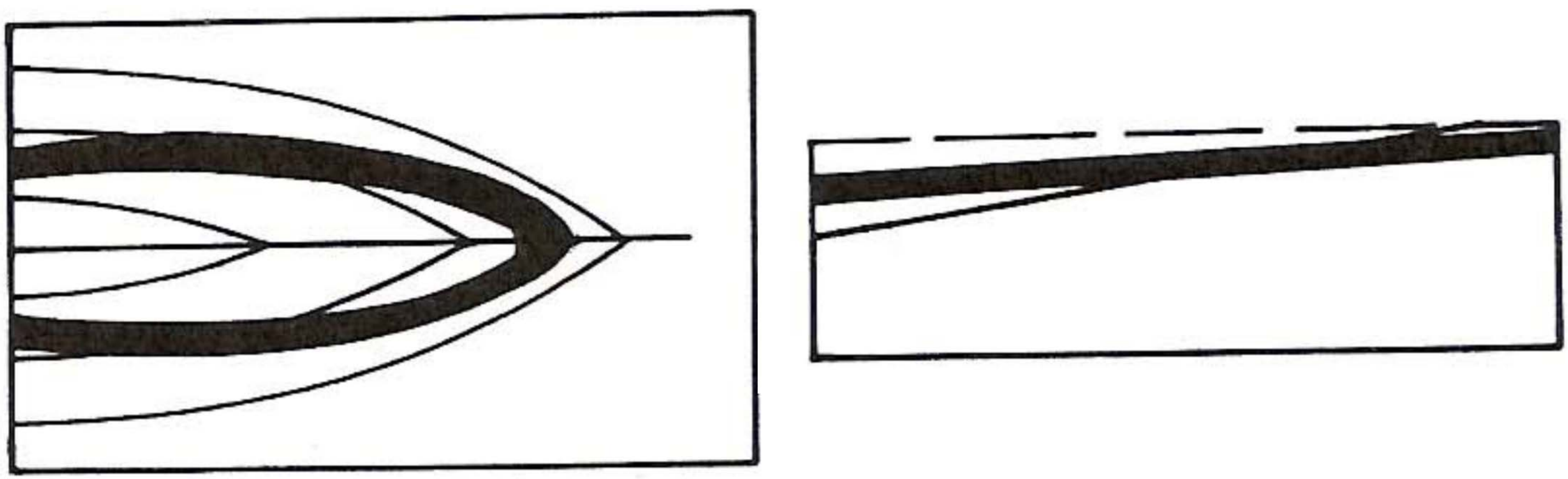
plane dipping downstream at an angle equal to the stream gradient;

Case 4a: Plane dipping upstream with an angle = stream gradient



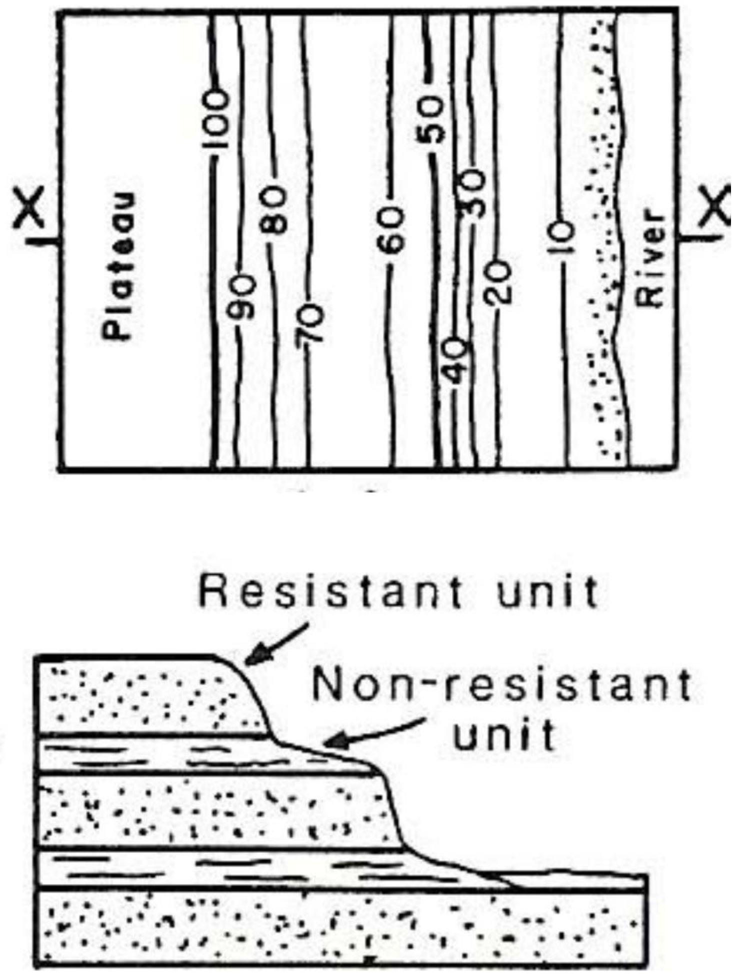
No V shaped outcrop formation by the strata.Bed/strata runs as two beds on two opposite walls of the valley without meeting each other further down stream plane dipping down stream at an angle less than the stream gradient.

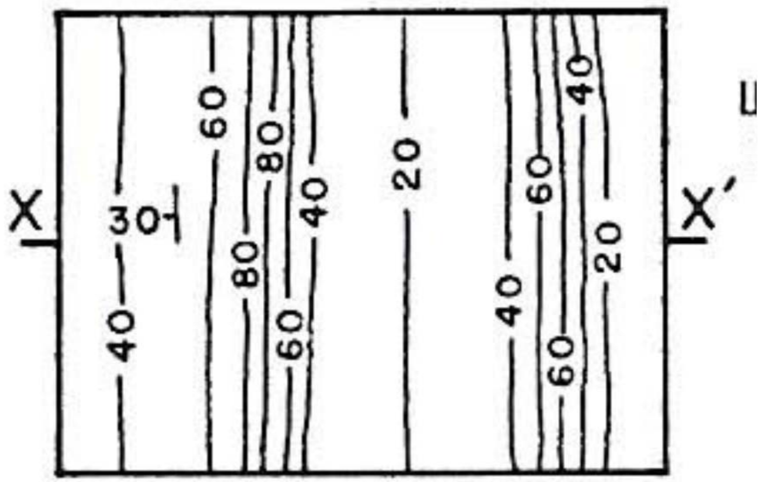
Case 4a: Plane dipping upstream with an angle < stream gradient

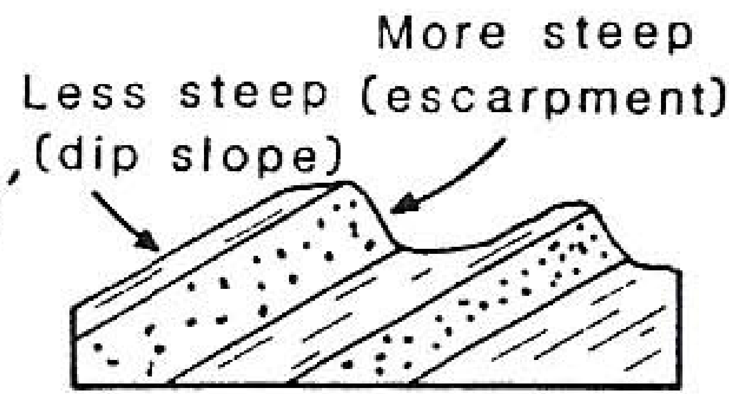


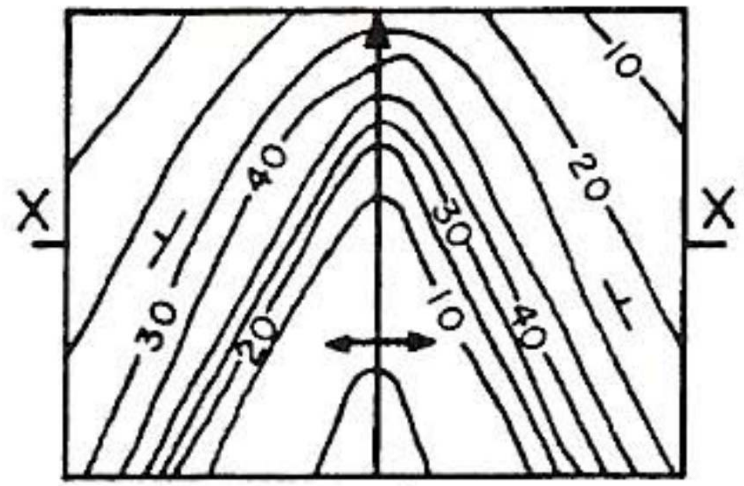
Topographic V and Bed V in same direction. Wings of v's tappers Towards each other.

Interpretation of topographic maps

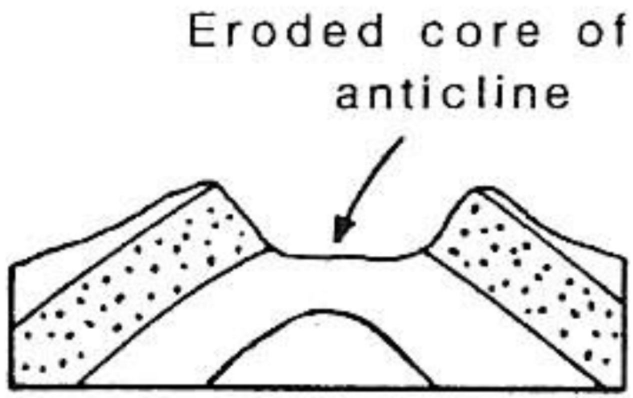


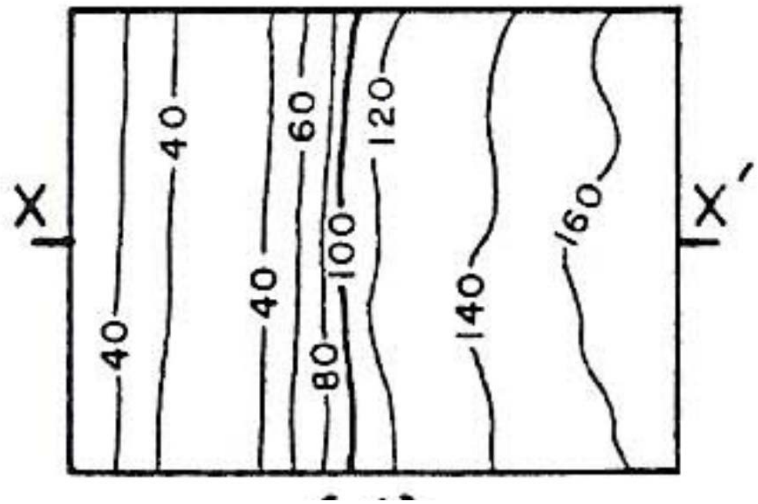

X
X'
30
40
60
80
80
60
40
20
40
60
60
40
20

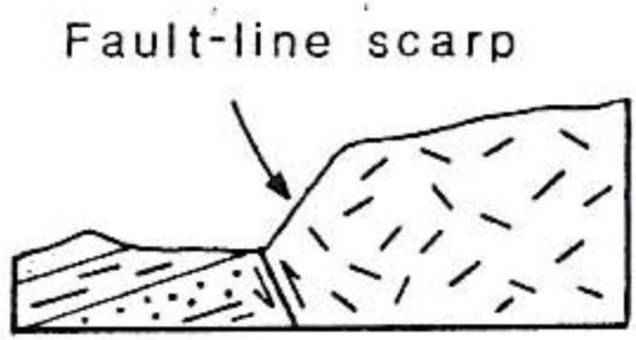

More steep
(escarpment)
Less steep
(dip slope)



(c)




X
X'
40
40
60
40
80
100
120
140
160


Fault-line scarp

Harrisburg
0 8
km
CI equal 100'